

中山大學
SUN YAT-SEN UNIVERSITY



国家超级计算广州中心
NATIONAL SUPERCOMPUTER CENTER IN GUANGZHOU

计算机图形学

发展史及应用

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课程概要

- 现代计算机图形学
- 计算机图形学发展史
- 计算机图形学应用
- 如何学习计算机图形学

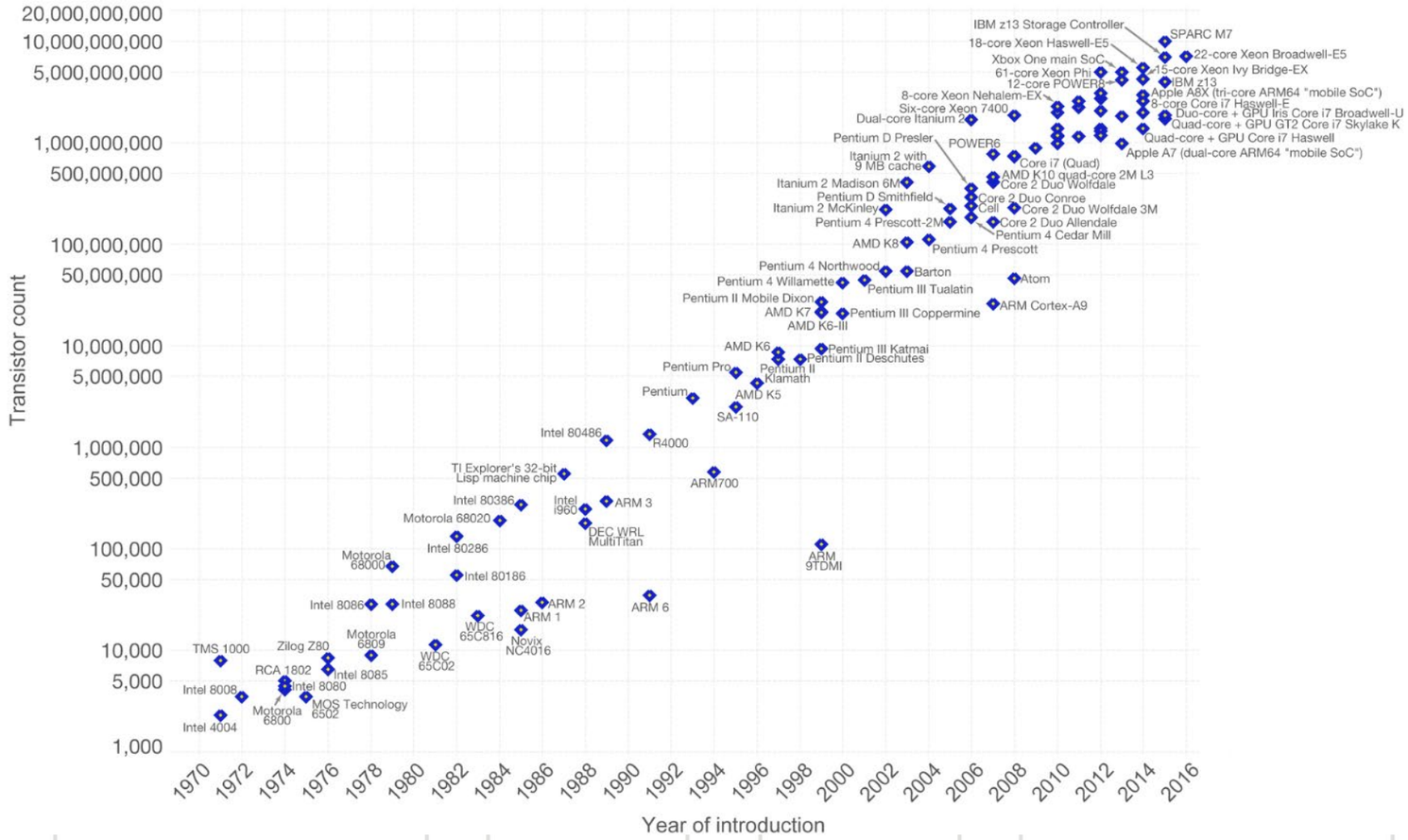
- 硬件变革:显卡运算性能的爆炸性增长
  - 模型处理及图形渲染需要大量运算
  - 摩尔定律

英特尔创始人之一戈登·摩尔提出：

“集成电路上可容纳的晶体管数目，约每隔两年便会增长一倍”

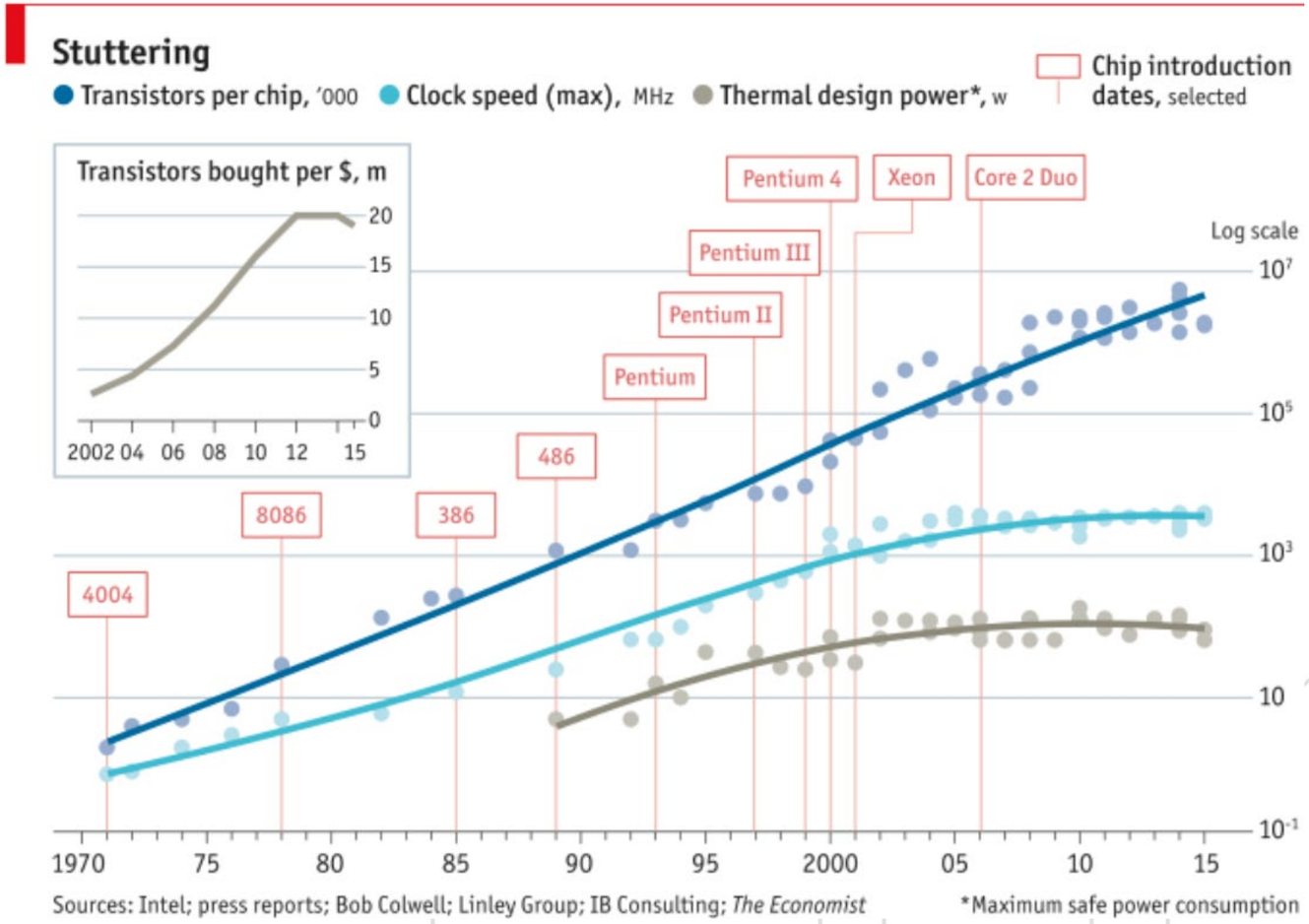
另一版本为18个月（由英特尔首席执行官大卫·豪斯提出）

- 硬件变革:显卡运算性能的爆炸性增长
  - 摩尔定律



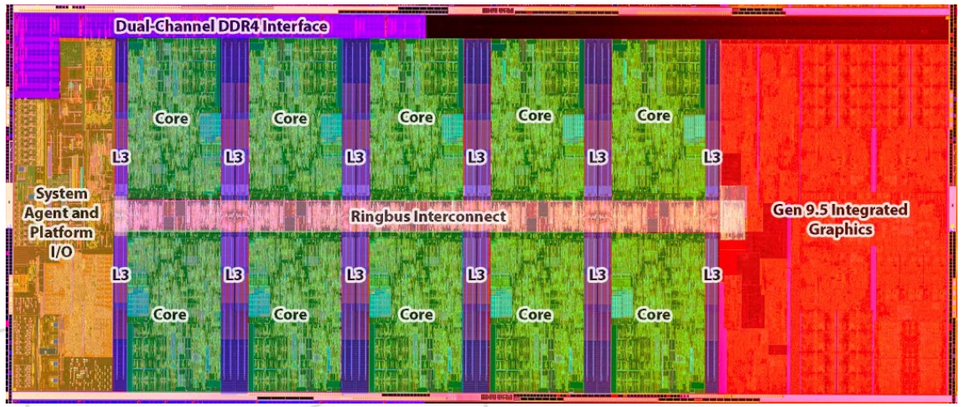
Licensed under CC-BY-SA by the author Max Roser.

- 硬件变革:显卡运算性能的爆炸性增长
  - 摩尔定律的瓶颈：晶体管数目≠频率



摩尔定律还能持续多久？

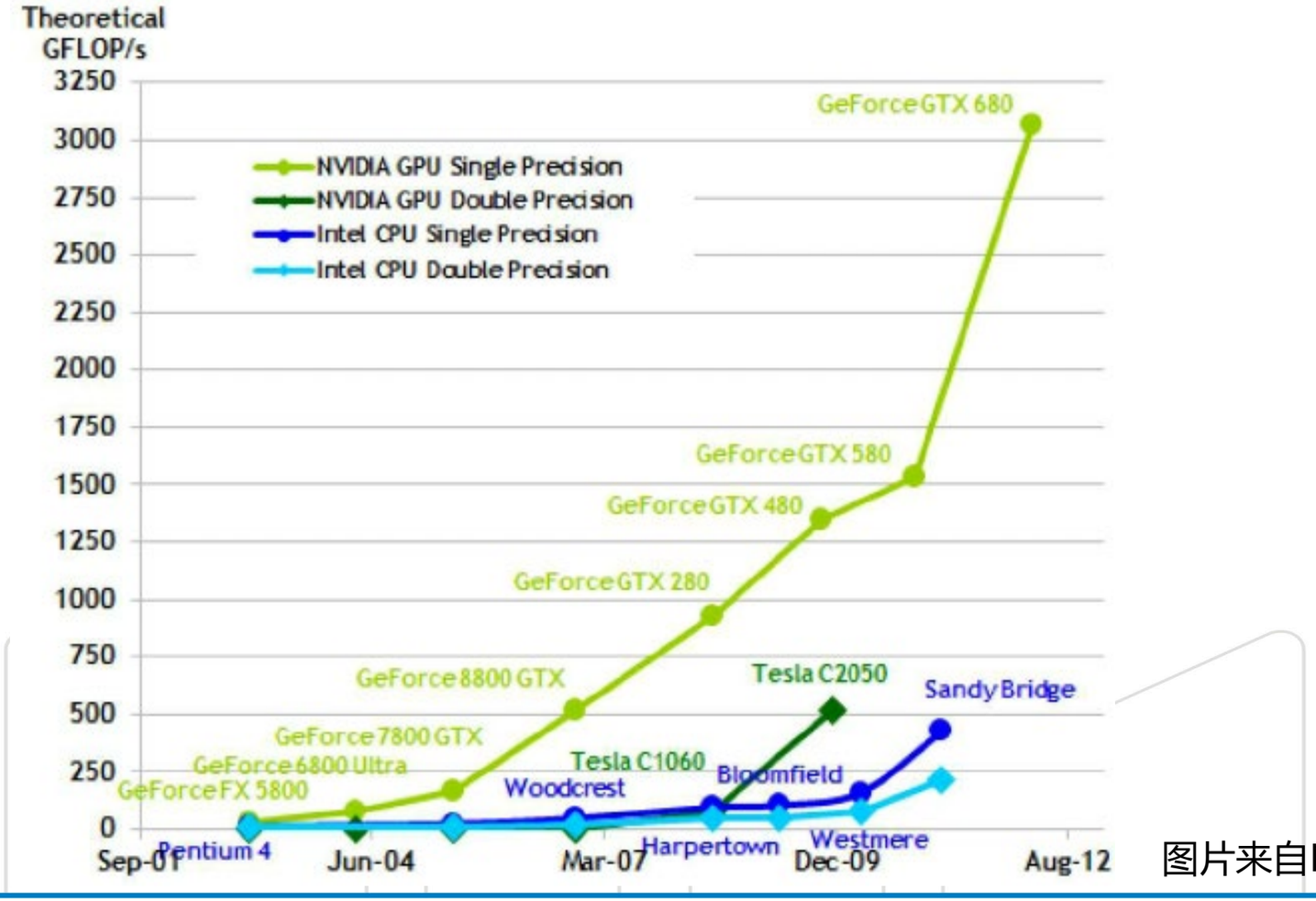
- 2021/2025年假说
- 量子穿隧效应



图片来源：TechPowerUp

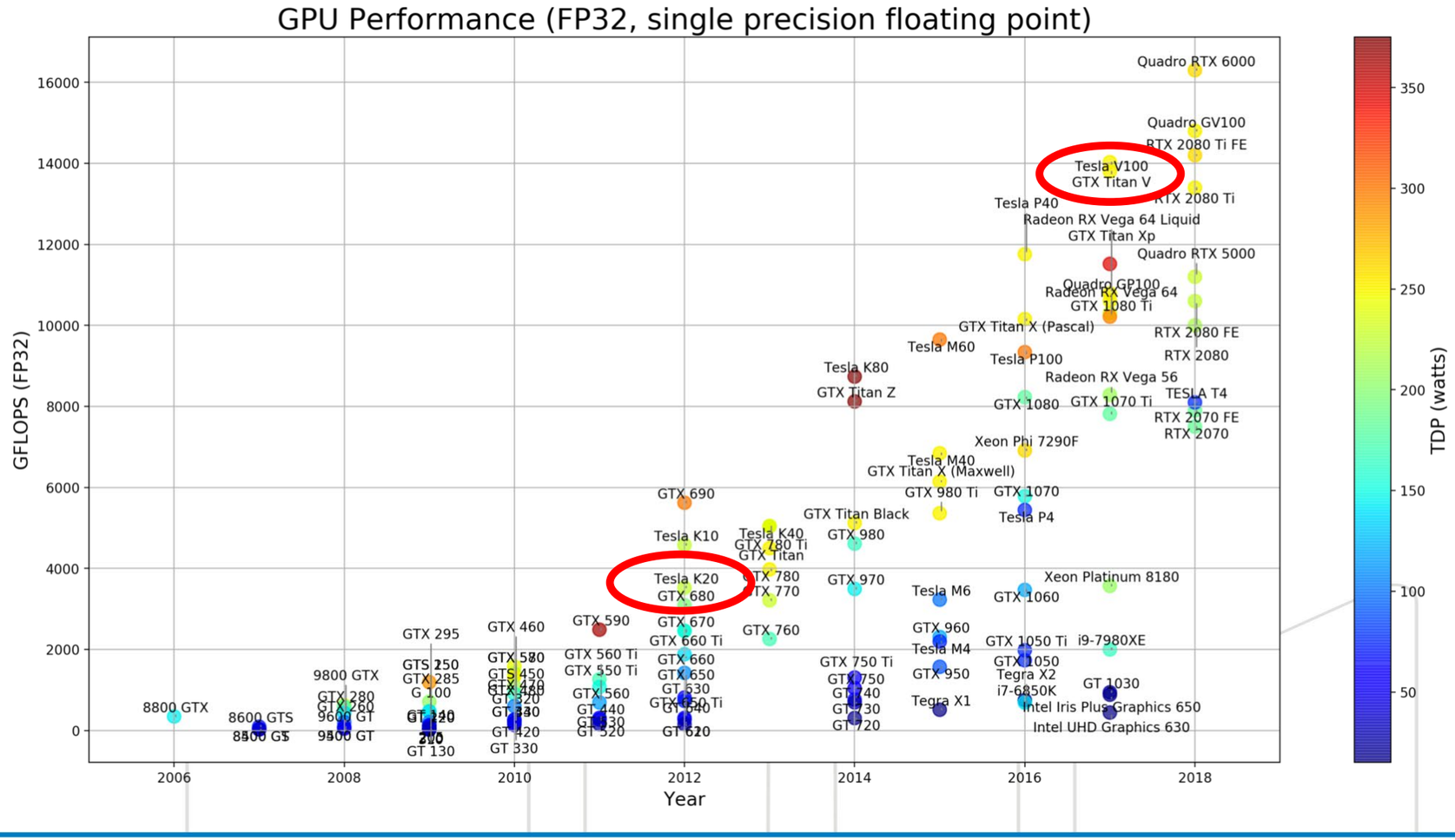
图片来源：The Economist’s Technology Quartlerly

- 硬件变革:显卡运算性能的爆炸性增长
  - CPU与GPU性能对比

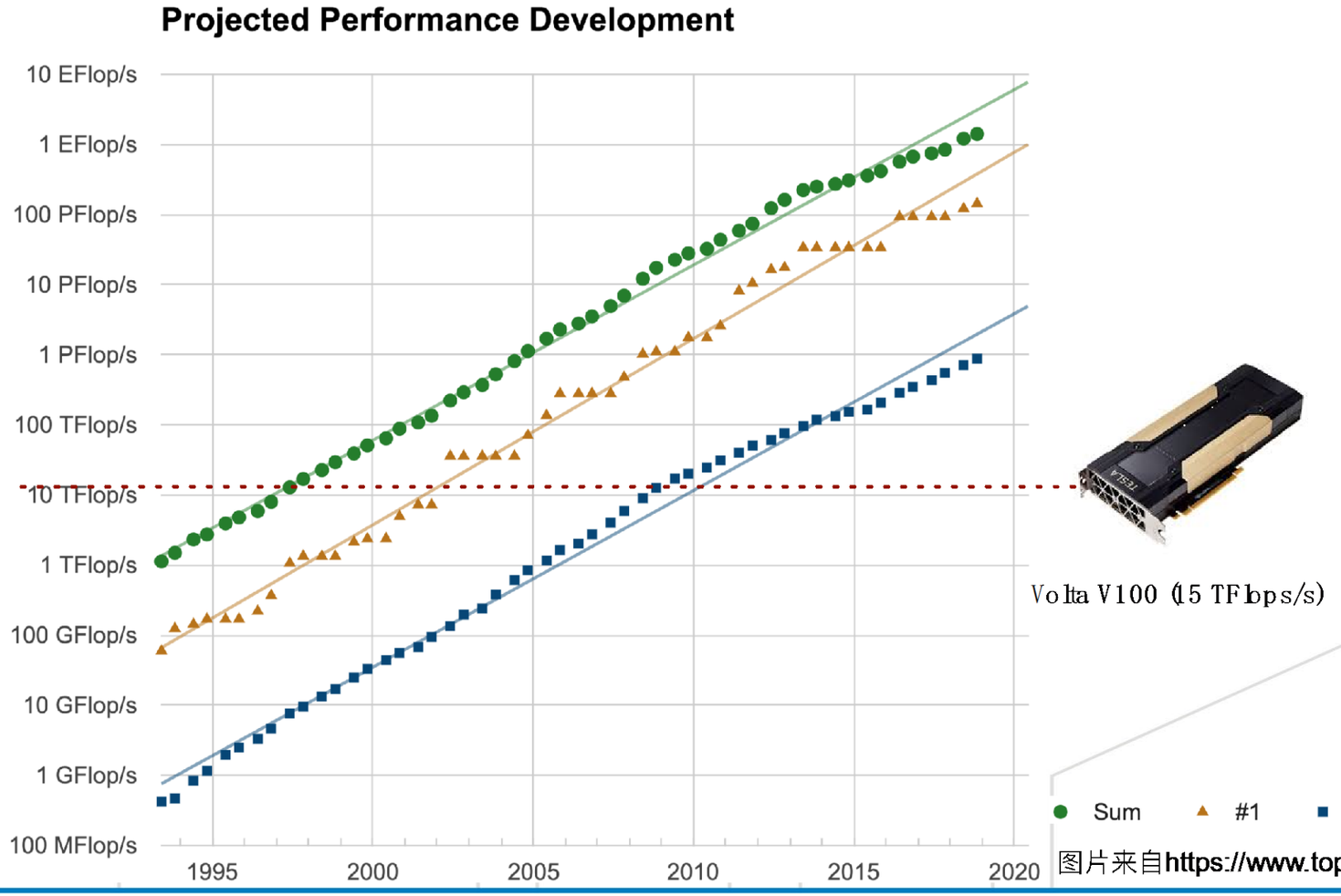


图片来自NVIDIA

- 硬件变革:显卡运算性能的爆炸性增长
  - GPU运算速度发展



- 硬件变革:显卡运算性能的爆炸性增长
  - GPU与超级计算机

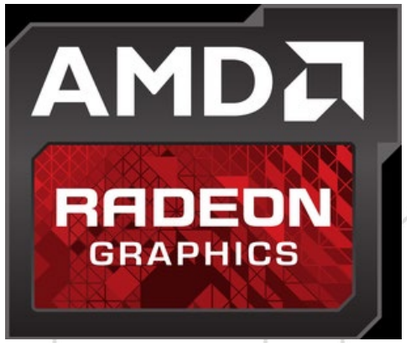


Volta V100 (15 TFlops/s)

图片来自https://www.top500.org/statistics/perfdevel/

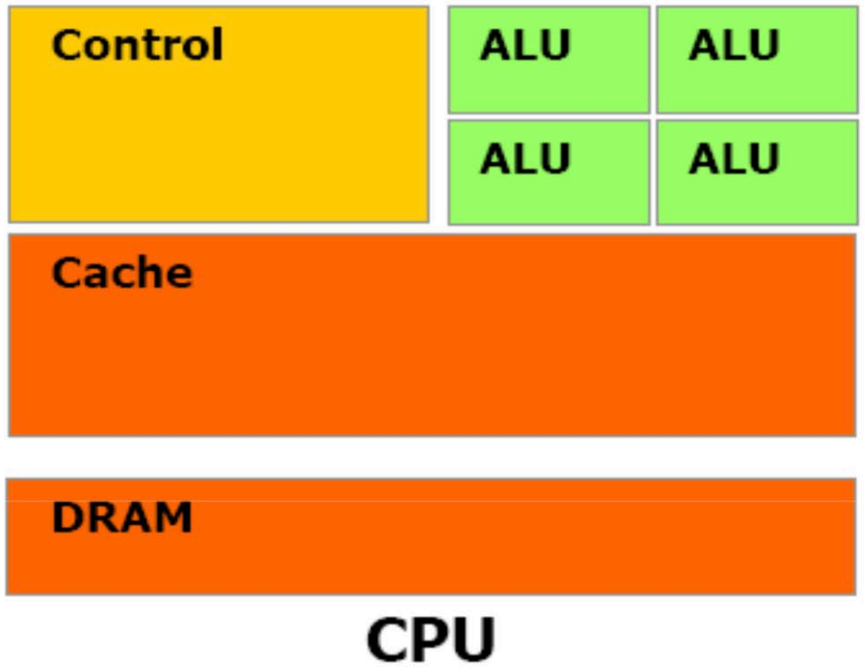
- 硬件变革:显卡运算性能的爆炸性增长：起源
  - 将图形处理从CPU转移至为其特殊设计的芯片上
  - NVidia GeForce™, AMD Radeon™, and Intel HD and Iris Pro Graphics
  - GPU最初只用于处理特定的图形学相关计算（渲染）
  - 当前，GPU越来越多地用于其他类型的通用计算
    - GPGPU (General-Purpose computing on the Graphics Processing Unit)

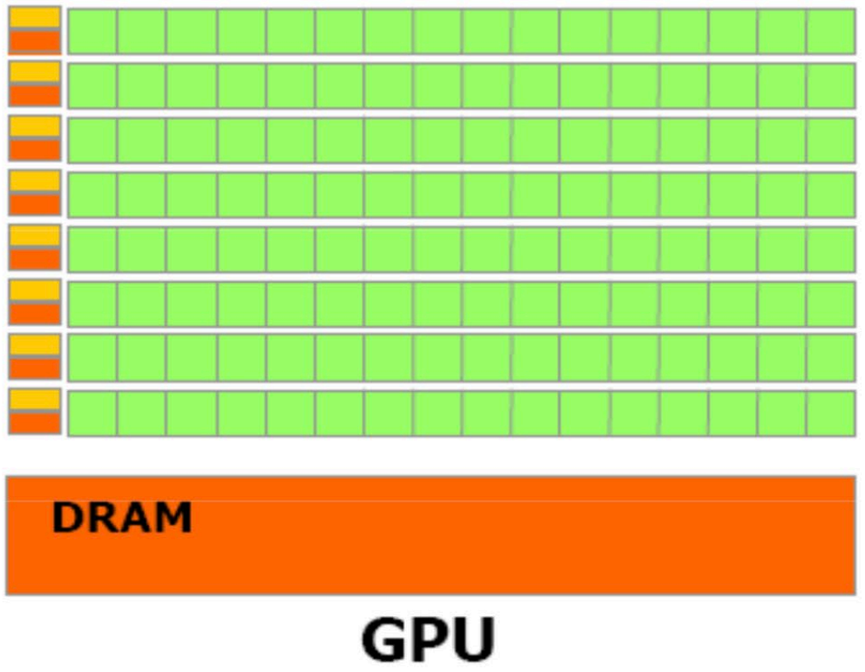






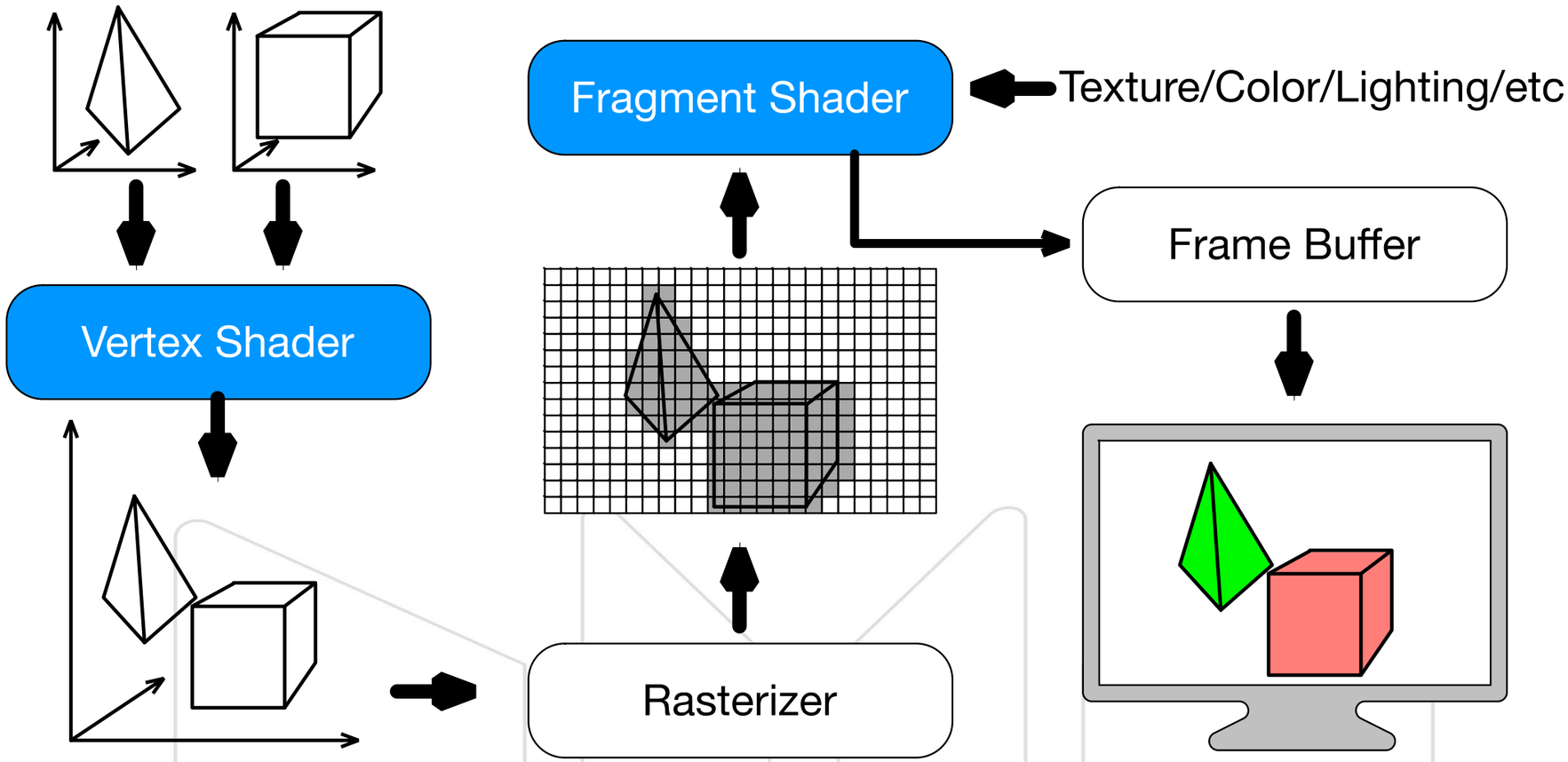
- 硬件变革:显卡运算性能的爆炸性增长
  - GPGPU
    - 高度并行化：相同的指令应用于大量数据上
    - 大量运算核心



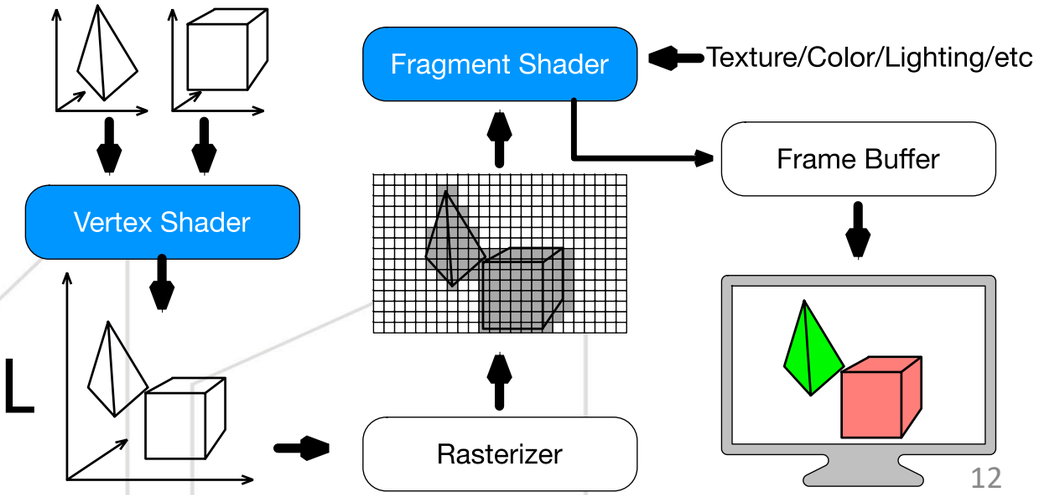


图片来自NVIDIA

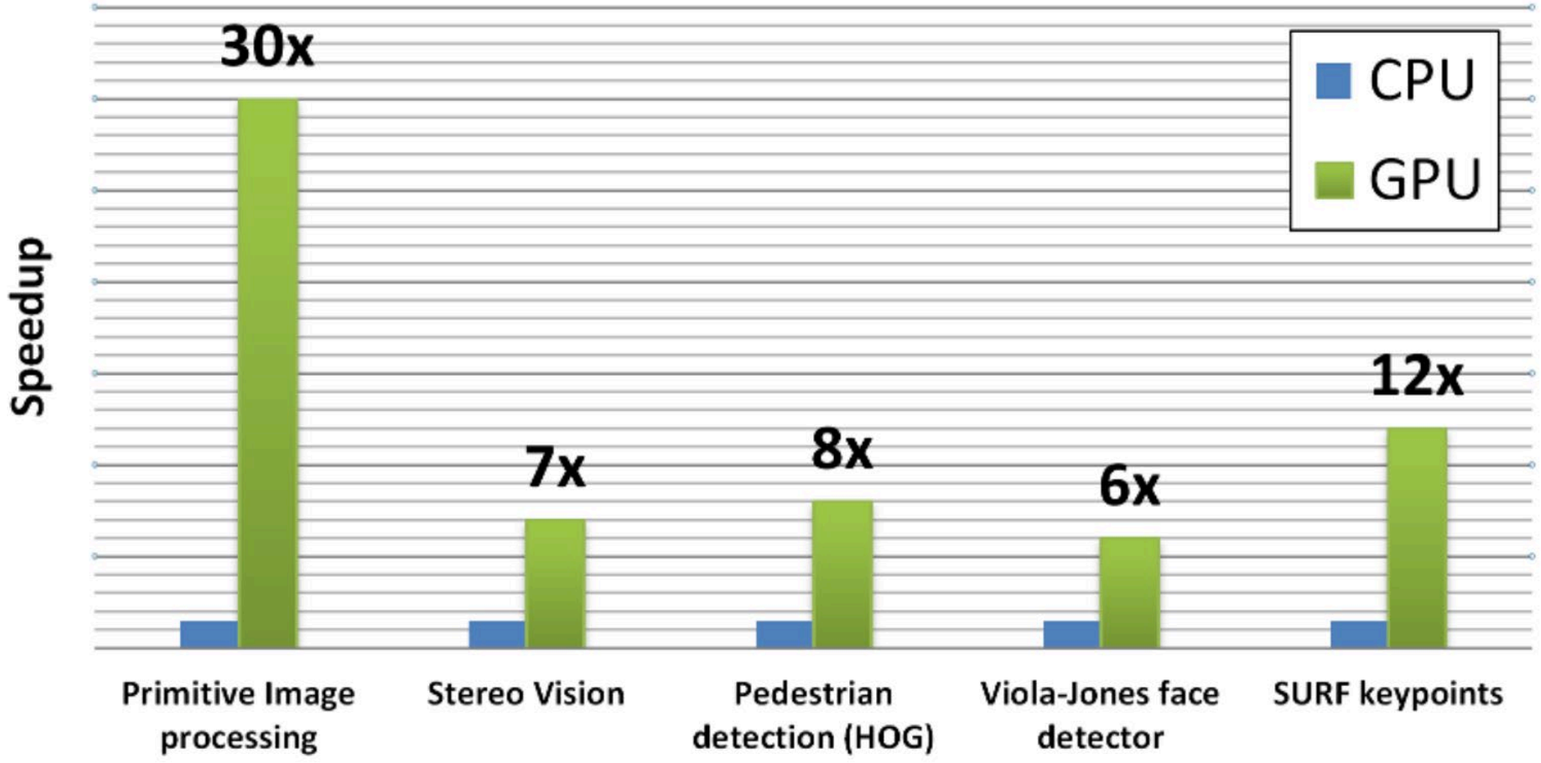
- 硬件变革:显卡运算性能的爆炸性增长
  - GPGPU
    - 图形绘制简易流程



- 硬件变革:显卡运算性能的爆炸性增长
  - GPGPU基本思路
  - Vertex shader 和 fragment shader 可由用户自行通过shading language编程
  - 把封装数据封装成图形绘制库所需的数据形式传入
  - 绘制过程中通过修改shader执行用户编写的程序
    - 通常是fragment shader
  - 将结果绘制到texture中传出
  - 需要很强的图形学编程能力
    - 出现专用工具，如CUDA、OpenCL



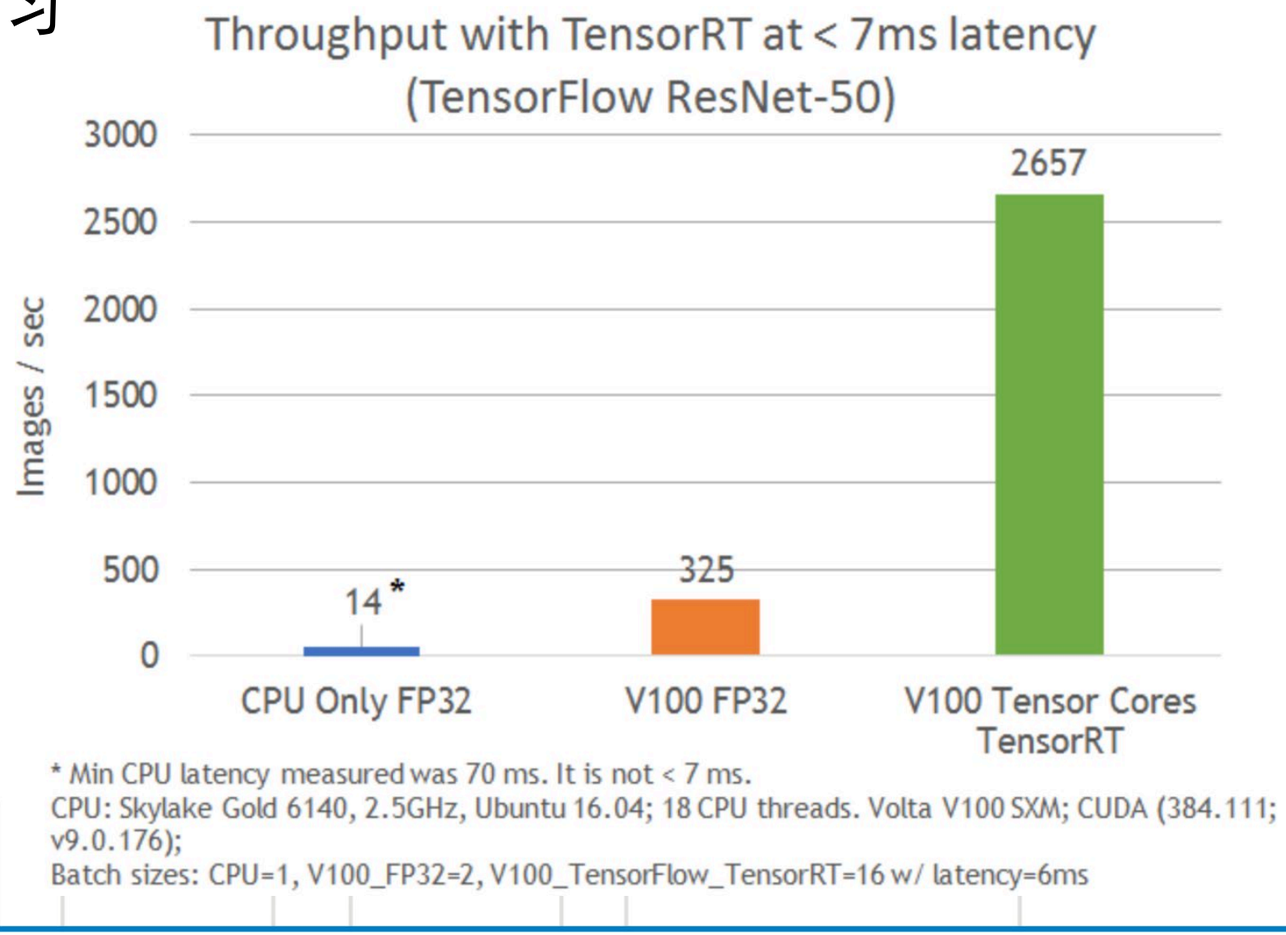
- 硬件变革:显卡运算性能的爆炸性增长
  - GPGPU加速效果
    - OpenCV——跨平台计算机视觉库



图片来自OpenCV.org

Tesla C2050 (448 cores) versus Core i5-760 2.8Ghz

- 硬件变革:显卡运算性能的爆炸性增长
  - GPGPU加速效果
    - TensorFlow——深度学习



图片来自Laurence Moroney

- 硬件变革:显卡运算性能的爆炸性增长
  - PC上的高端独立显卡已逐渐取代专用图形工作站
  - 由高速总线或局域网连接的PC集群常用于PowerWall、Caves等大规模图形显示
  - PC集群也被应用于大规模计算，如，蛋白质折叠、天气预报等
  - 如今多GPU也可以通过NVIDIA SLI (Scalable Link Interface)桥接



SGI图形工作站



多GPU通过SLI桥接

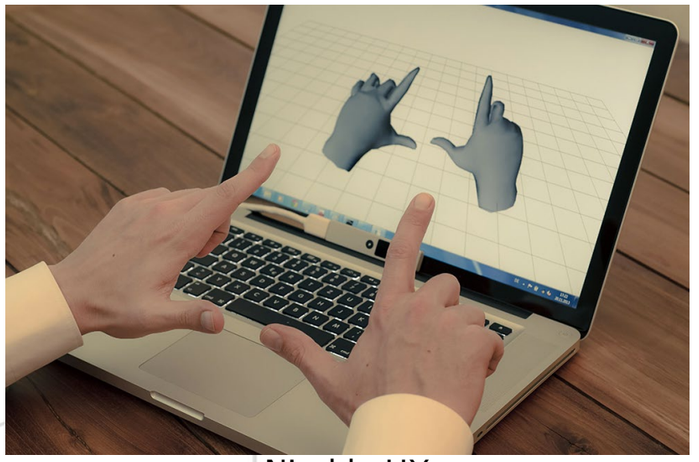
硬件变革:输入设备丰富

- 鼠标、平板电脑与手写笔、多点触控、力反馈、游戏控制器、扫描仪、数码相机（图像，计算机视觉）
- 体感交互设备
  - Xbox Kinect: “you are the controller”



Xbox Kinect


Leap Motion


Nimble UX

- 硬件变革:输出设备丰富
  - 智能手机/笔记本电脑/台式电脑/平板电脑
  - 智能手表
  - 头戴式显示（Head-Mounted Display, HMD)
  - 3D沉浸式虚拟现实


Apple iPhone

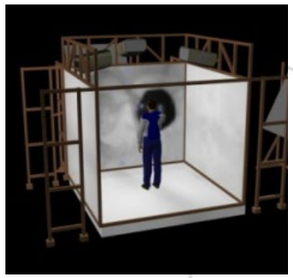

Android Phones


Microsoft's first Surface


Tablets


Apple Watch


Android Wear


Brown's old Cave


Microsoft Hololens


Oculus Rift


Google Cardboard

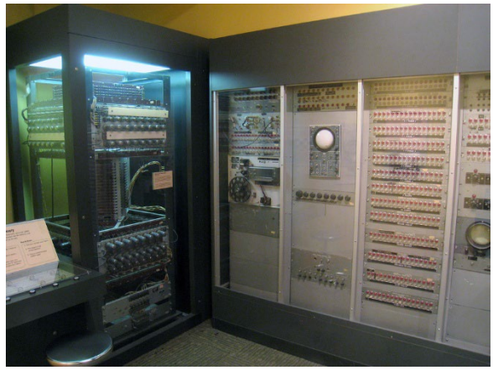
- 软件变革
  - 算法与数据结构
    - 建模算法：重现各种三维物体的形状、材质
    - 渲染算法：产生自然的效果
    - 数据结构：如用于加速光线追踪的几何算法
    - 神经网络渲染：神经网络拟合数据+传统体数据渲染
  - 并行化
    - 计算机图形学中的大多数操作都为embarrassingly parallel
      - 如，在渲染时计算像素颜色往往与其他像素无关
  - 分布式与云计算
    - 将操作发送至云端执行，隐藏执行过程直接获取执行结果
    - 渲染甚至可以作为网络服务提供给用户
      - 如，超算上通过科学模拟产生的数据的可视化

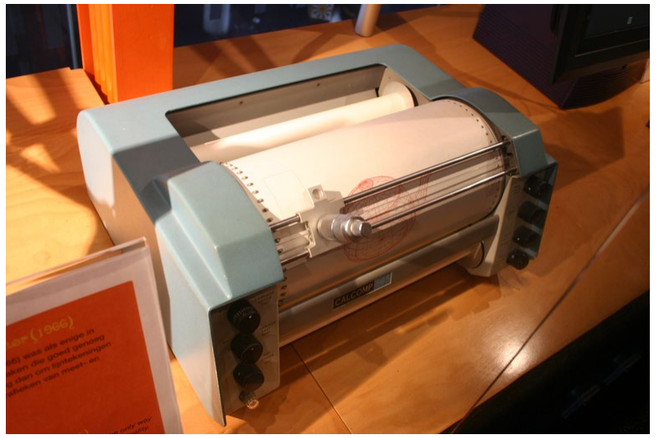
课程概要

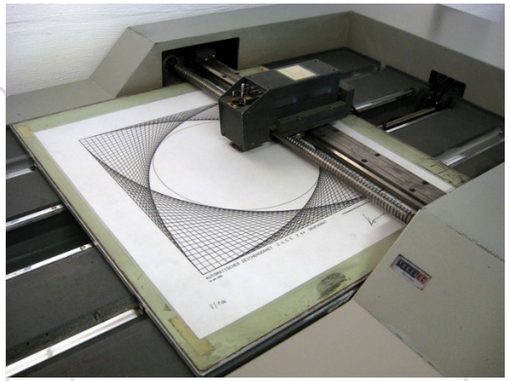
- 现代计算机图形学
- 计算机图形学发展史
- 计算机图形学应用
- 如何学习计算机图形学

计算机图形学发展史

- 50年代
  - 1951年，MIT的Jay Forrester和Robert Everett为旋风一号（Whirlwind I）设计了第一个可视显示单元（VDU）
  - 1958年，CalComp开发了565滚筒绘图仪（drum plotter）
  - 1958年，Gerber公司开发了第一台平板绘图仪（flat plotter）







计算机图形学发展史

- 60年代
  - Spacewar！（星级飞行）为最早的电脑游戏
    - MIT学生Steve Russell在DEC PDP-1计算机上开发
    - 两个玩家各控制一艘星际战舰对战

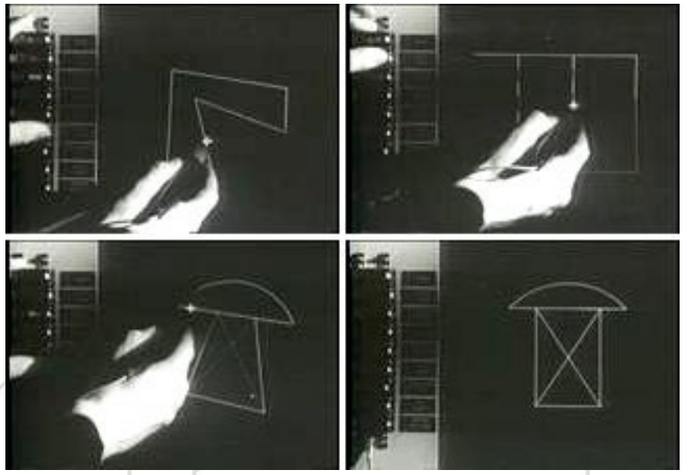




计算机图形学发展史

- 60年代
  - 1963年，交互式系统Sketchpad (Robot Draftsman) 诞生
    - Ivan Sutherland博士论文
      - BS, 卡内基理工学院；MS, Caltech；PhD, MIT
    - 1988年图灵奖

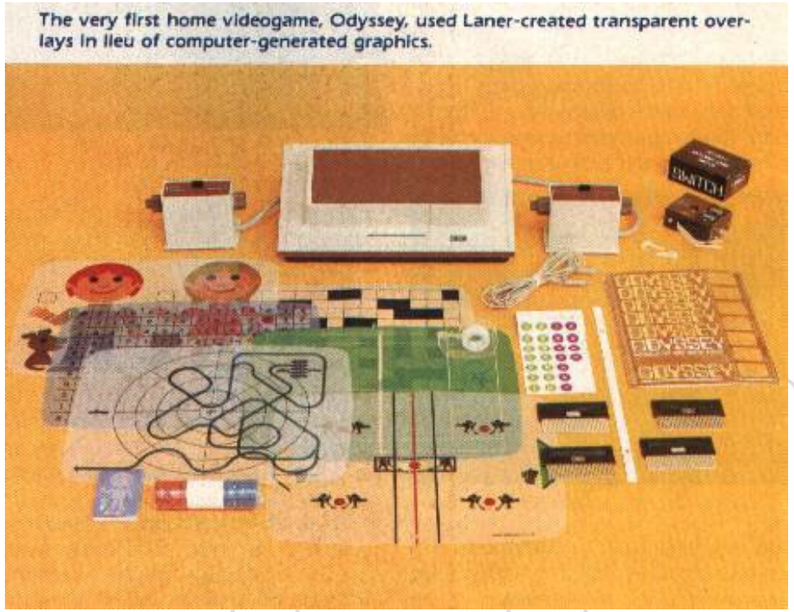




计算机图形学发展史

- 60年代
  - 第一台商业家庭视频游戏终端Magnavox Odyssey诞生
    - Ralph H. Baer设计，工程师William Harrison和William Rusch协助开发
    - 1966年开始，1968年产生第一个原型

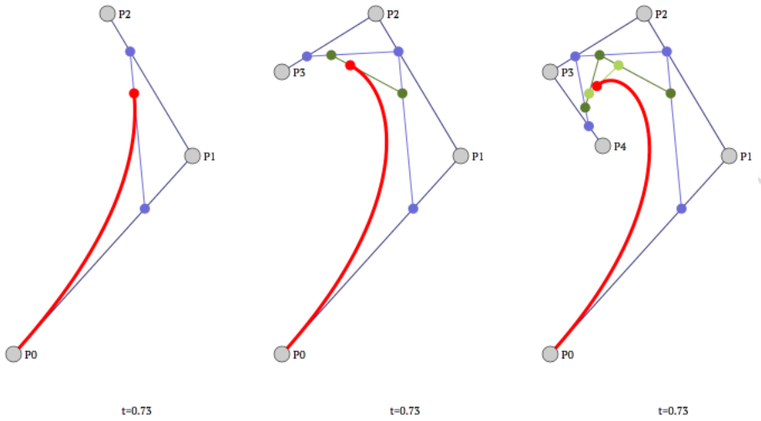






计算机图形学发展史

- 60年代
  - 计算机辅助设计（Computer Aided Design, CAD)诞生
    - 1958年，MIT教授Steven Anson Coons提出CAD概念
    - 1964年，提出Coons surface
    - 1985年，ACM SIGGRAPH的Steven A. Coons奖因其命名
  - 60年代末，法国工程师Pierre Bezier创造Bezier曲线和曲面
    - Bezier曲面成为当今CAD应用最广的曲面之一



图片作者：Jason Davis

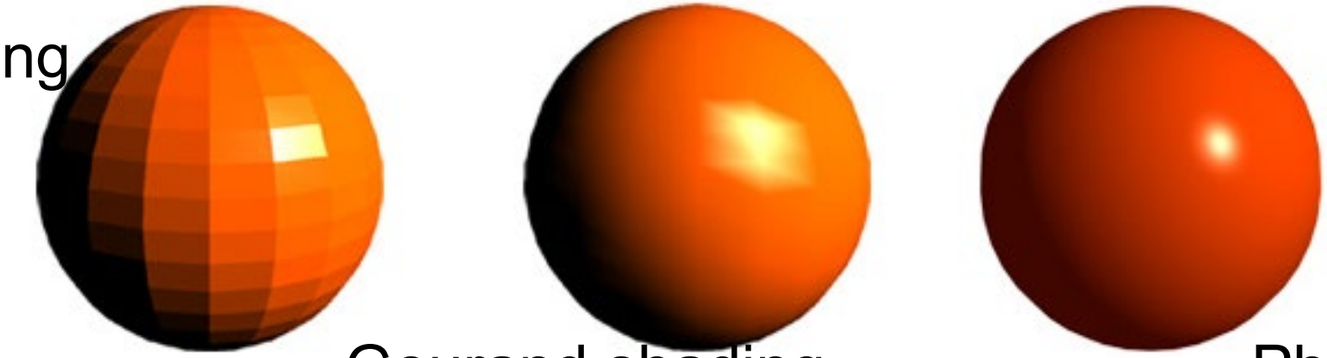
计算机图形学发展史

- 70年代
  - 光栅化图形算法快速发展
    - 扫描转化（scan conversion），裁减（clipping），surface hidden removal（消隐），及其他相关算法出现
  - 标准化进程
    - 1974年，ACM SIGGRAPH（Special Interest Group on computer GRAPHics and Interactive Techniques）成立图形标准委员会
    - 国际标准组织（ISO）发布CGI (Computer Graphics Interface), CGM (Computer Graphics Metafile), GKS (Graphics Kernel System), PHIGS (Programmer's Hierarchical Interactive Graphics Standard)

计算机图形学发展史

- 70年代
  - 1970年，Bouknight提出第一个光反射模型（flat shading）
  - 1971年，Henri Gouraud提出“漫反射（diffuse reflection）+插值（interpolation）”的反射模型（Gouraud shading）
  - 1975年， Bùi Tường Phong提出基于法向量插值的局部反射模型（Phong shading)，为当今最有影响力的模型
    - Utah大学
    - 73年博士毕业论文，75年发表论文

Flat shading

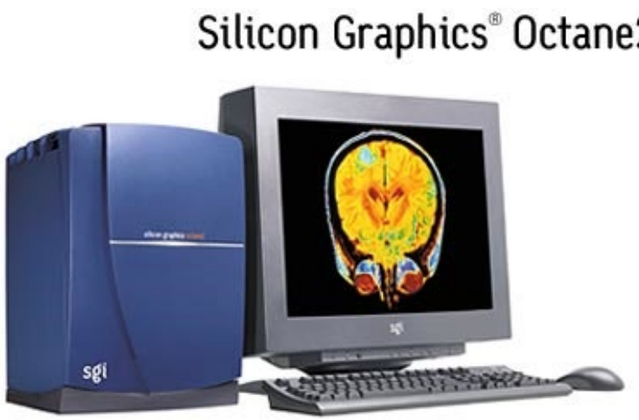


Gourand shading

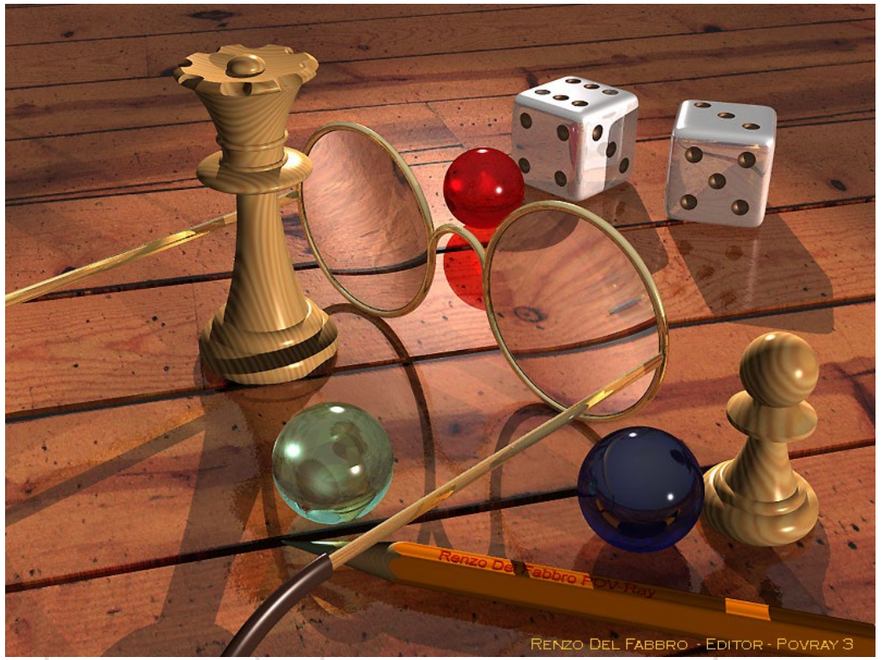
Phong shading

计算机图形学发展史

- 80年代
  - 光线追踪（ray tracing）和辐射度（radiosity）算法
    - Whitted提出光线追踪算法，考虑光的反射和透射
    - 计算机图形学里程碑的发展
  - 计算机图形硬件发展

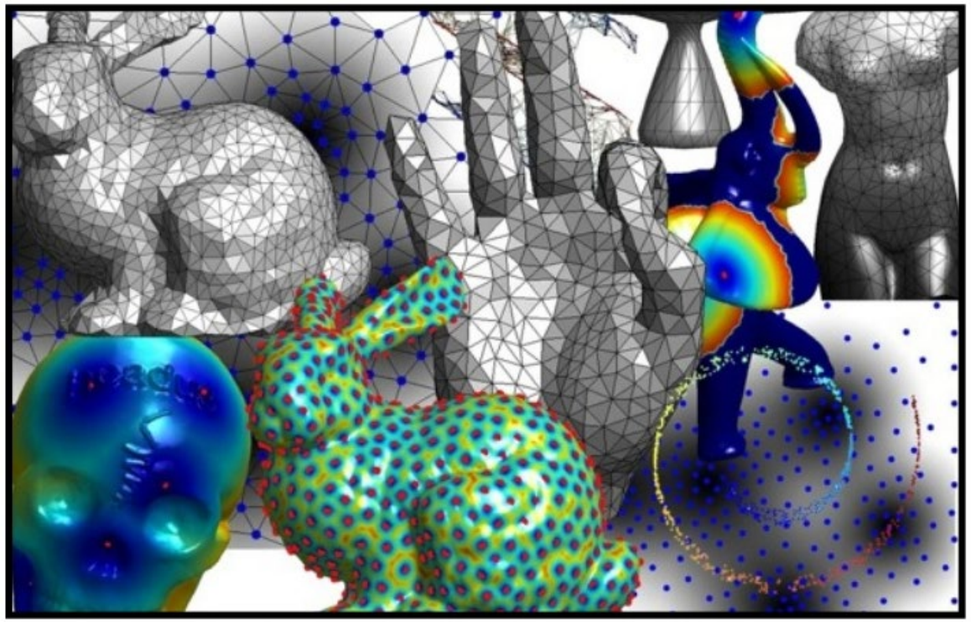


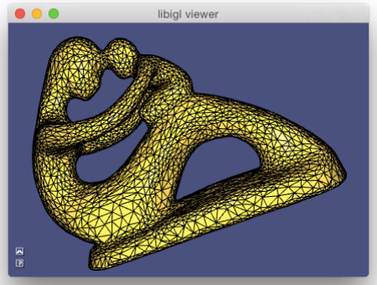
Graphics workstations such as these have been replaced with commodity hardware (CPU + GPU), e.g., MaxBuilts + Nvidia cards

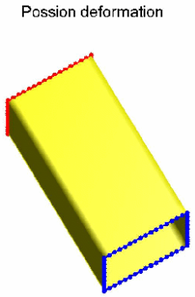


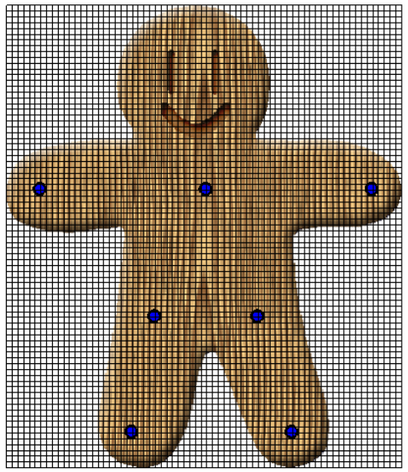
计算机图形学发展史

- 90年代
  - 建模
    - Meshes
    - Subdivision
    - Implicit Surface
    - Procedural
    - Multi-resolution
  - 渲染
    - Volume rendering
    - Image-based rendering
    - Point-based rendering



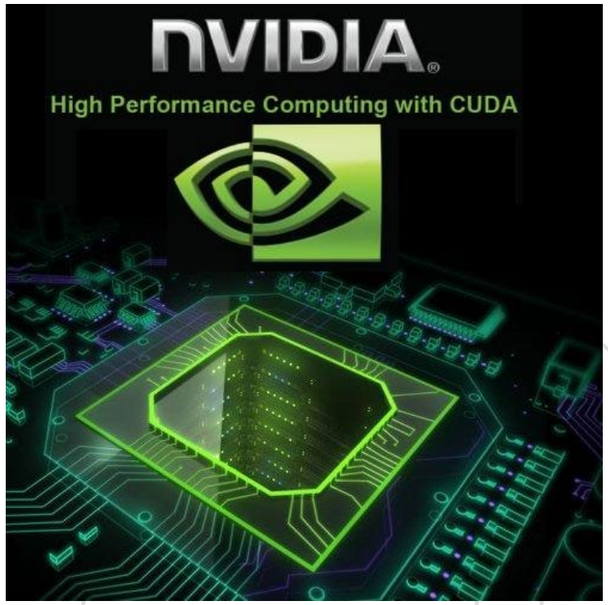


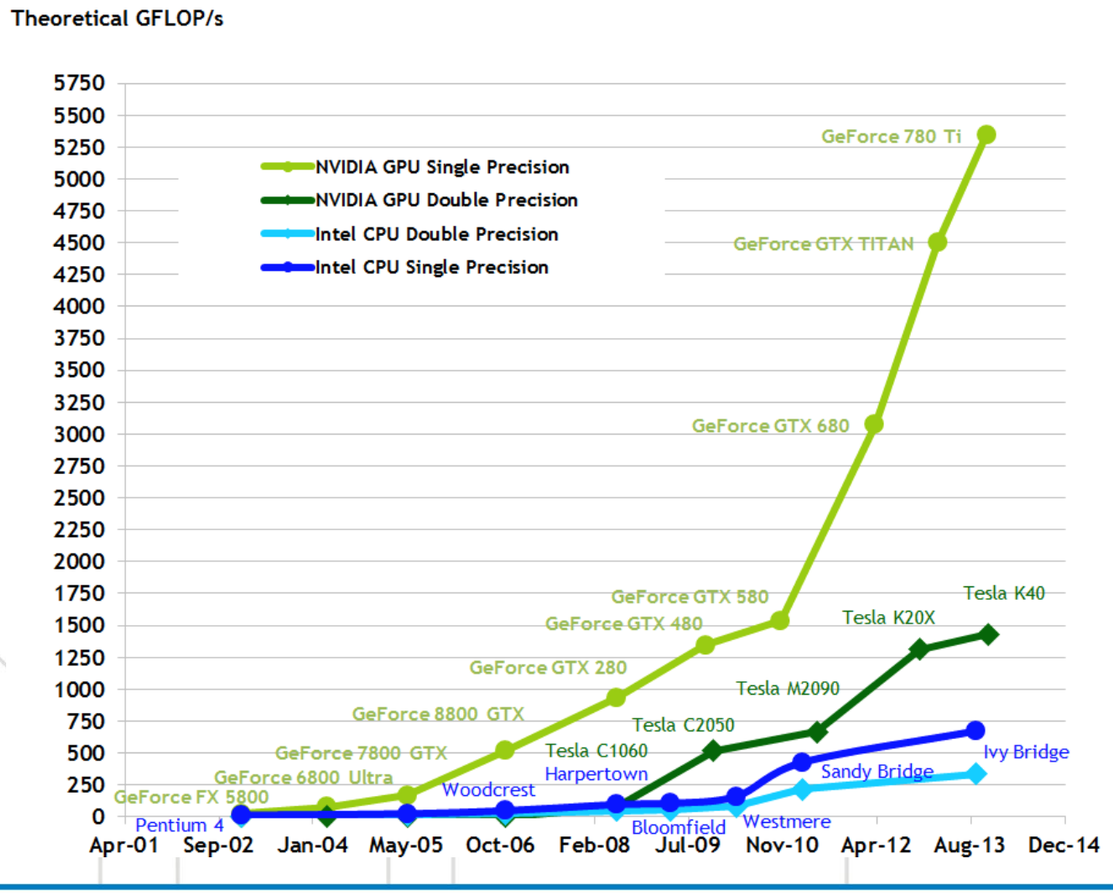




计算机图形学发展史

- 2000年后
  - 3D扫描技术
  - 图形硬件飞速发展
  - GPU并行编程
  - 深度学习兴起





计算机图形学发展史

- 新趋势
  - 体感设备：微软Kinects







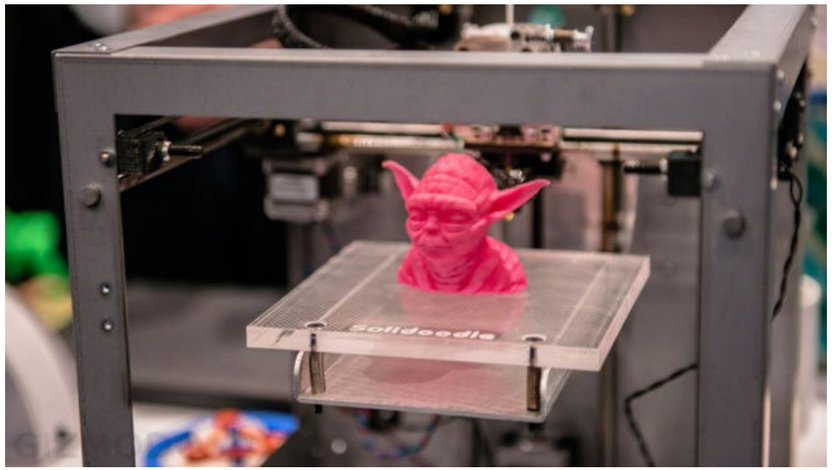




计算机图形学发展史

- 新趋势
  - 3D打印



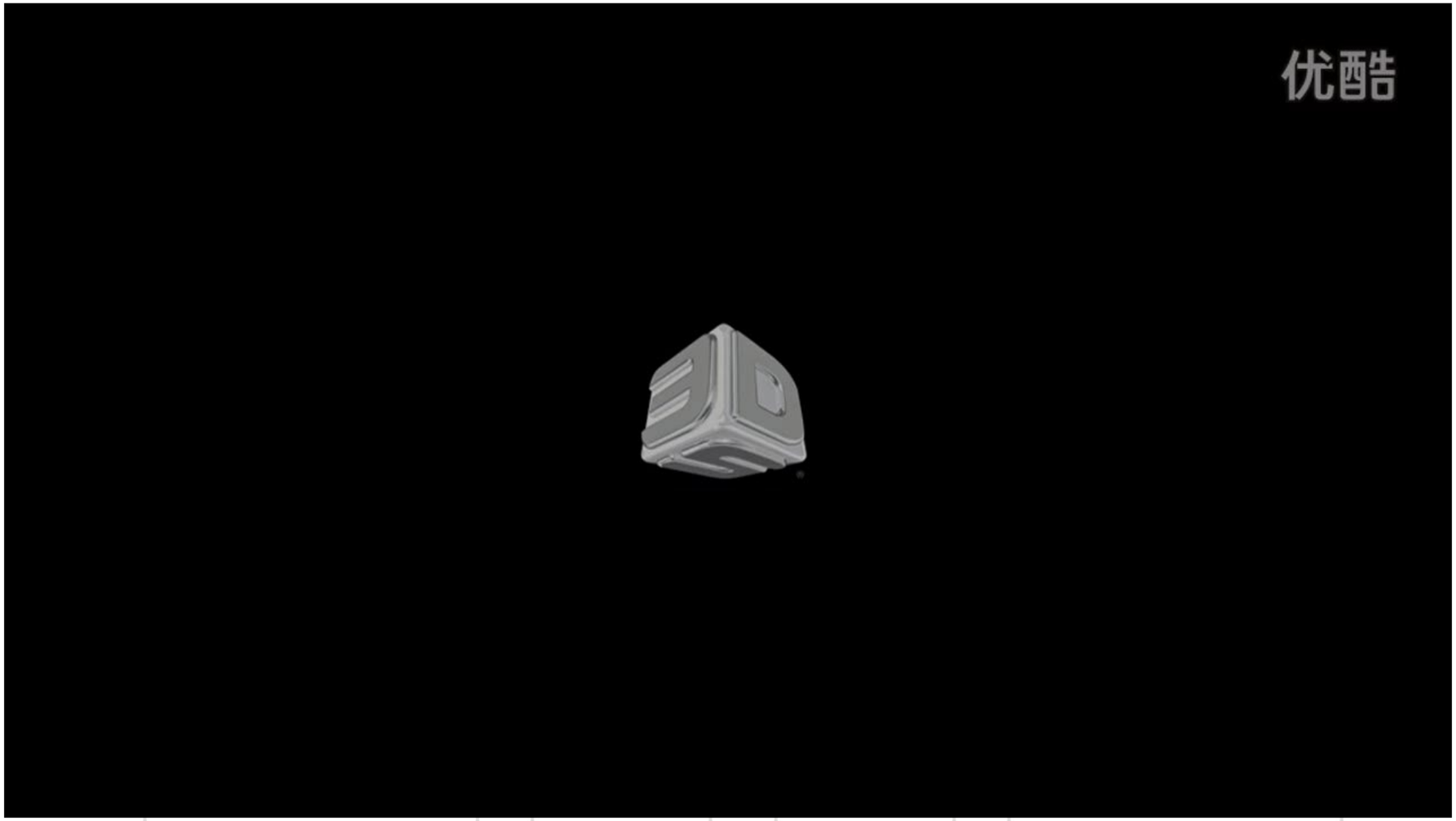






计算机图形学发展史

- 新趋势
  - 3D打印



计算机图形学发展史

- 新趋势
  - 新型交互设备：Leap Motion





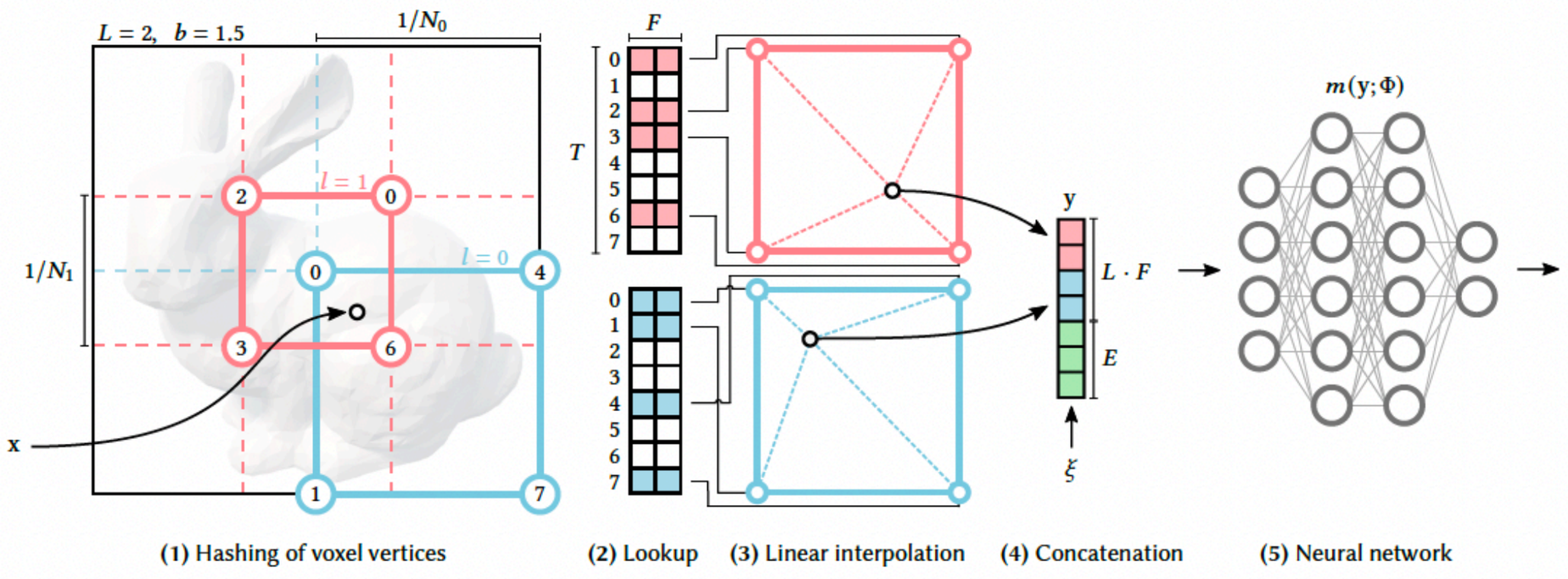


计算机图形学发展史

- 新趋势
  - 虚拟现实：Cyberith Virtualizer



- 新趋势
  - 神经辐射场（NeRF）



课程概要

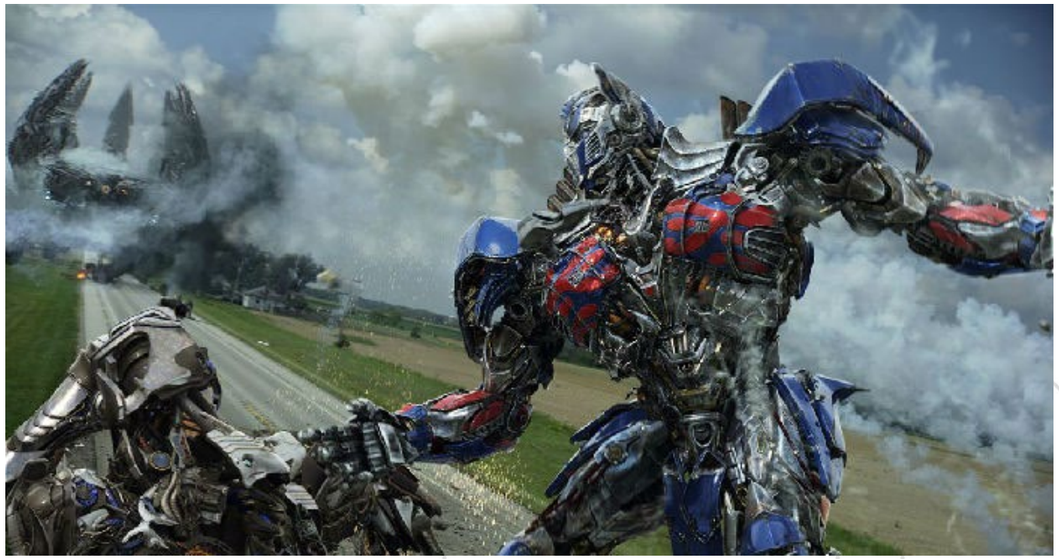
- 现代计算机图形学
- 计算机图形学发展史
- 计算机图形学应用
- 如何学习计算机图形学

计算机图形学应用

- 电影
  - 动画
  - 特效





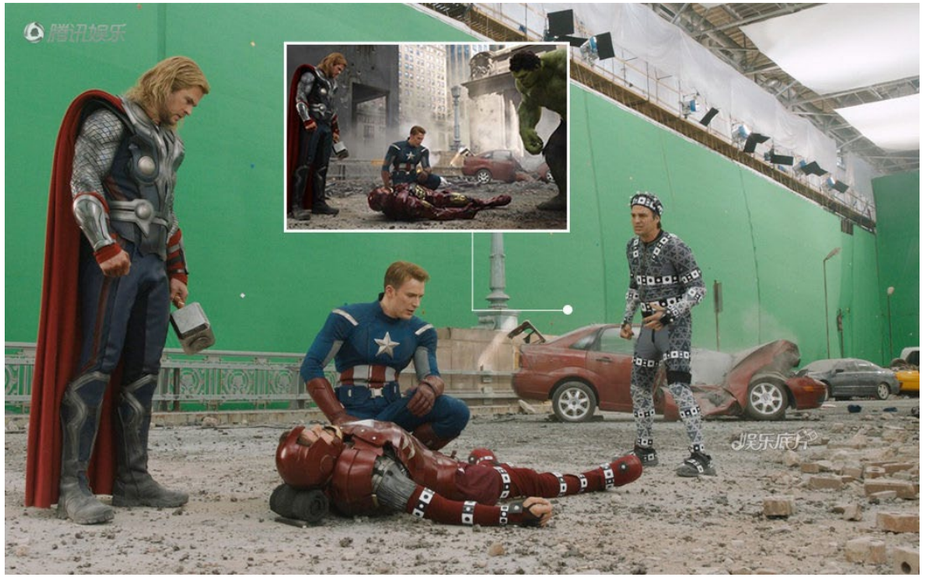




- 电影
  - 动作捕捉









- 游戏




KILLED BY
A tower was destroyed by The Dire!
LUNA
Respawning in 8 seconds
SHOP


6:07.14
BOUNTY > 59,750
MILESTONES > 3
5/35
BUSTED
EVADE
MPH


AC 100
HP 100
AMMO 42/210

计算机图形学应用

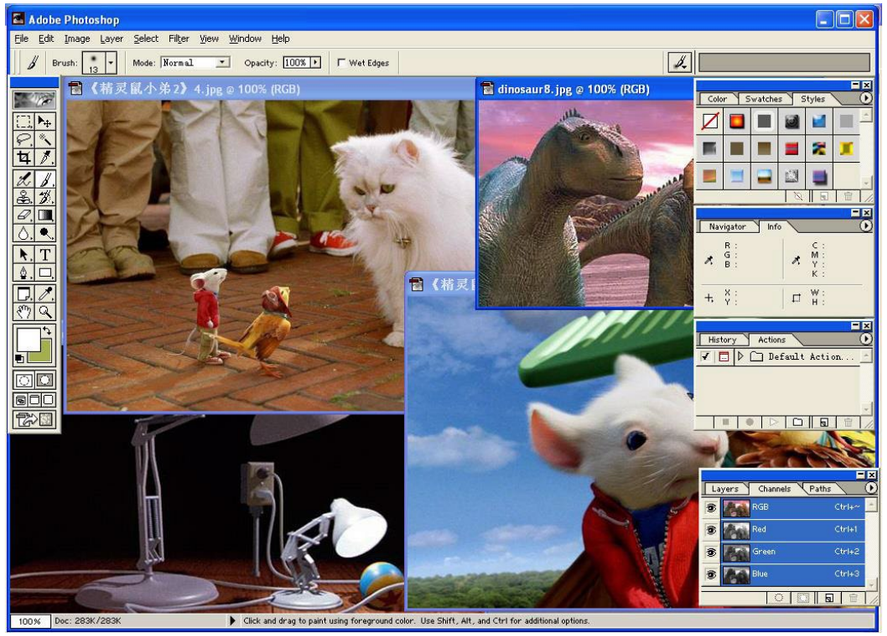
- 图像
  - 广告
  - 设计
  - 艺术



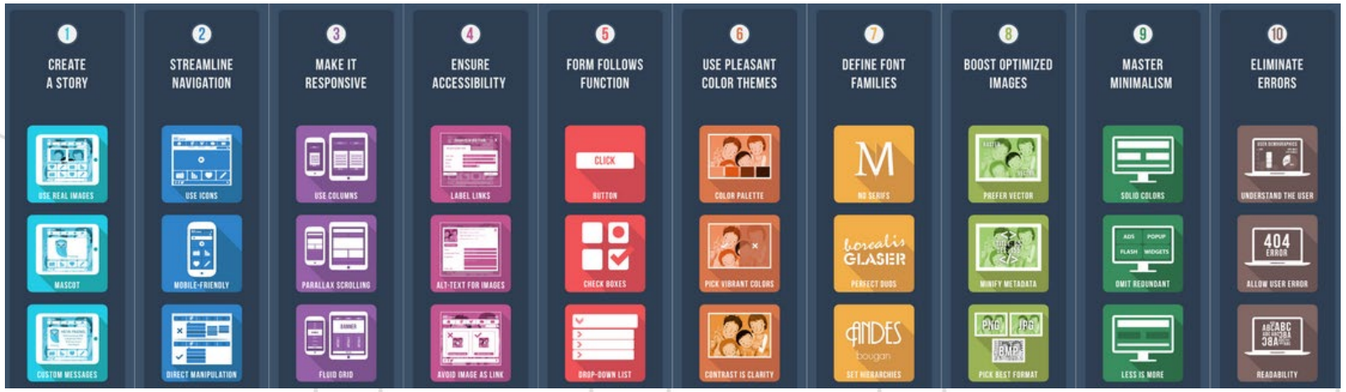




- UI设计


Adobe Photoshop
File Edit Image Layer Select Filter View Window Help
《精灵鼠小弟2》4.jpg @ 100% (RGB)
dinosaur8.jpg @ 100% (RGB)
Color Swatches Styles
Navigator Info
History Actions
Layers Channels Paths
RGB Ctrl+~
Red Ctrl+1
Green Ctrl+2
Blue Ctrl+3
100% Doc: 283K/283K
Click and drag to paint using foreground color. Use Shift, Alt, and Ctrl for additional options.


2:15
4:22
Great Big Momma Bird


1 CREATE A STORY
2 STREAMLINE NAVIGATION
3 MAKE IT RESPONSIVE
4 ENSURE ACCESSIBILITY
5 FORM FOLLOWS FUNCTION
6 USE PLEASANT COLOR THEMES
7 DEFINE FONT FAMILIES
8 BOOST OPTIMIZED IMAGES
9 MASTER MINIMALISM
10 ELIMINATE ERRORS

- 训练与模拟

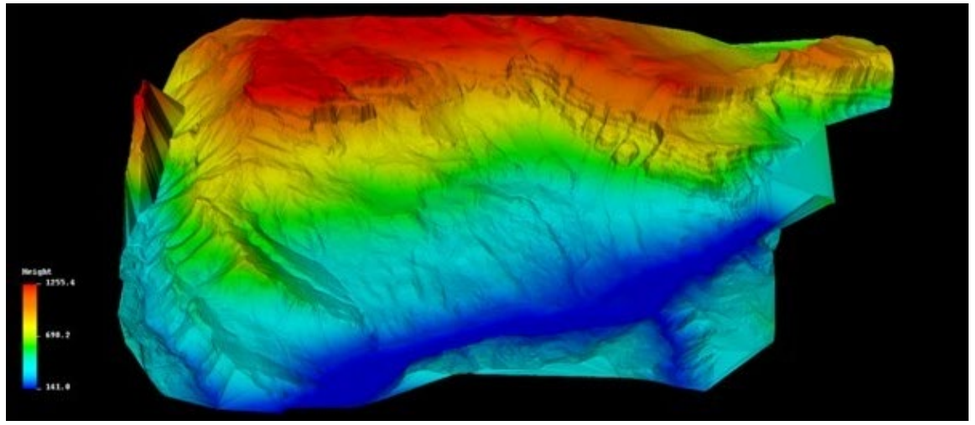




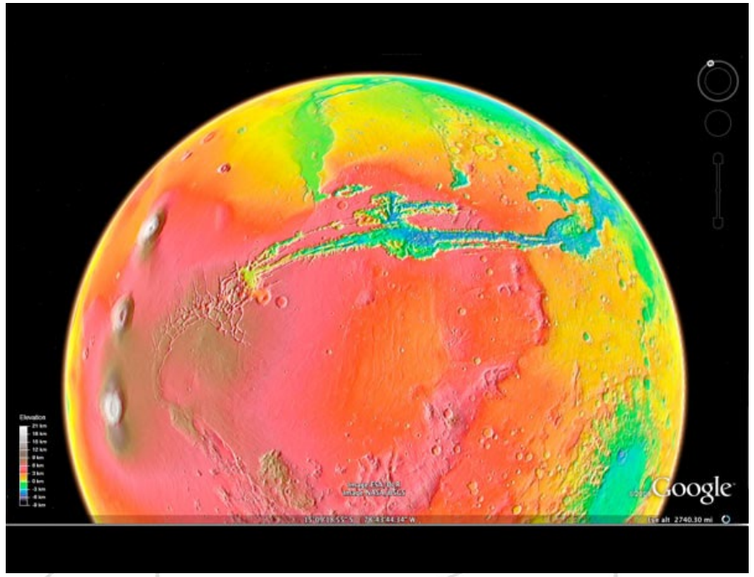




- 地理科学
  - 几何对齐算法、数字地球及数字城市

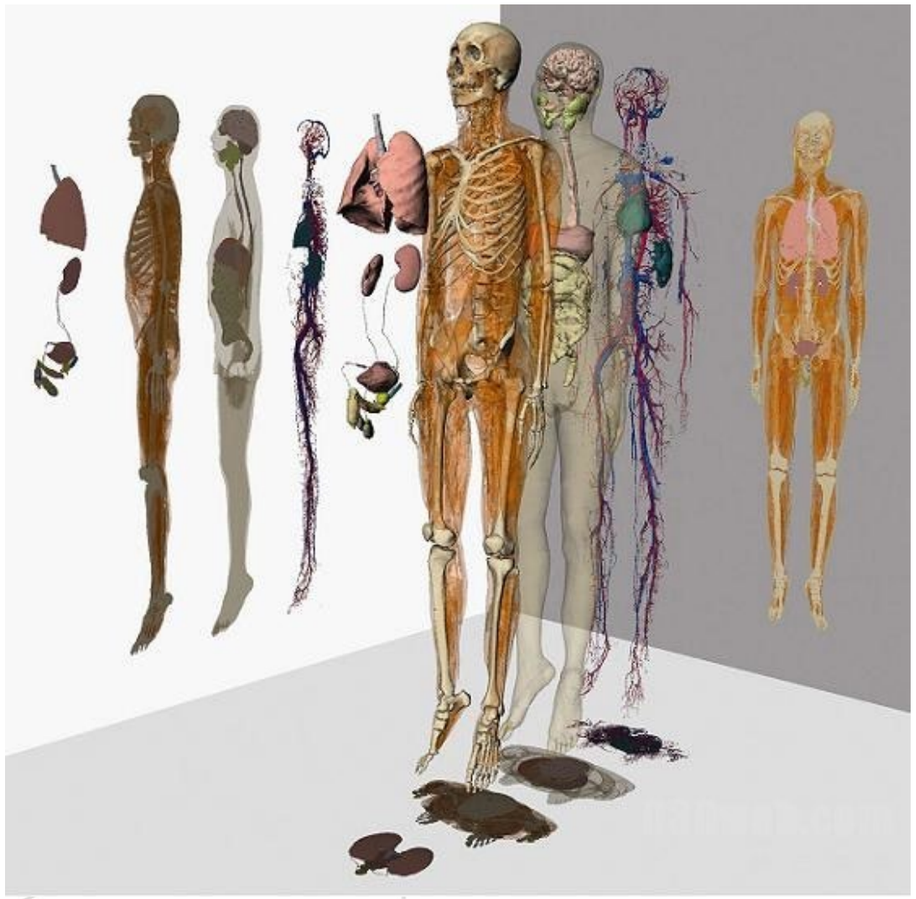




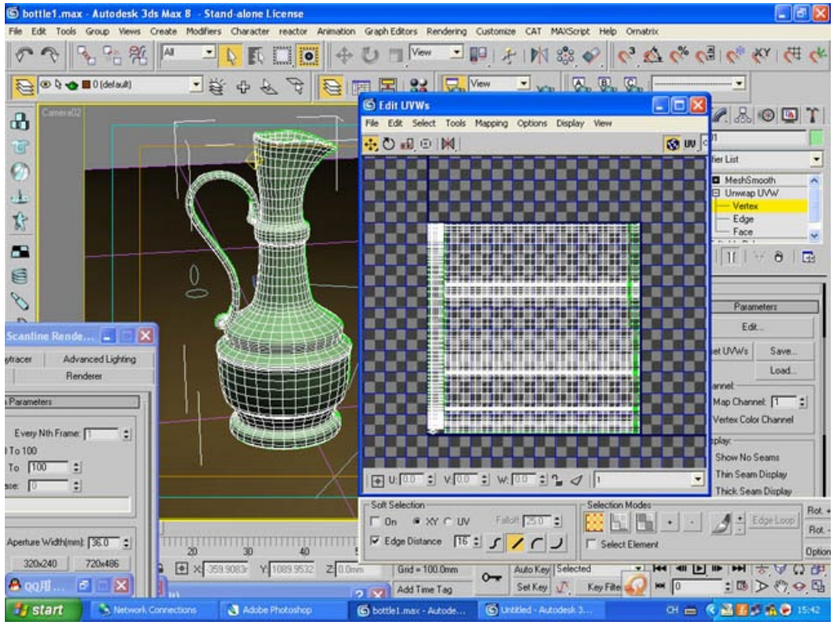


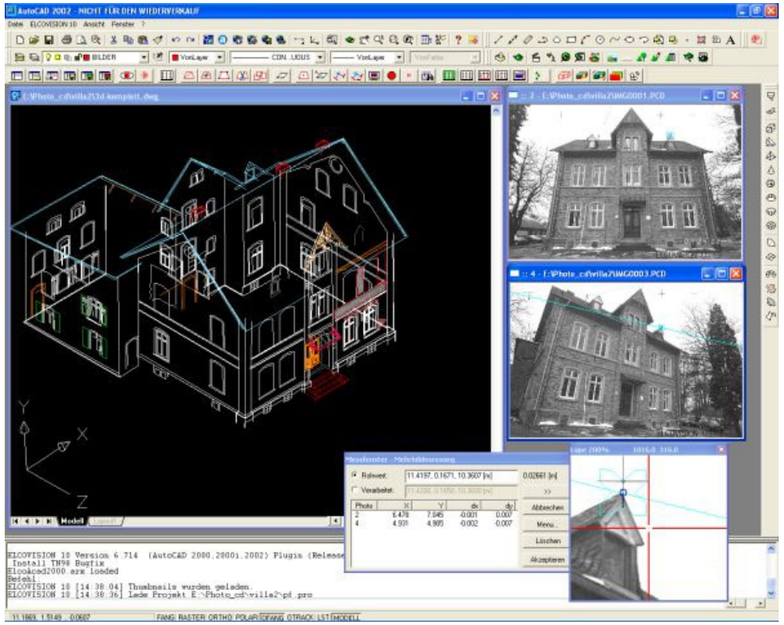
- 医学影像

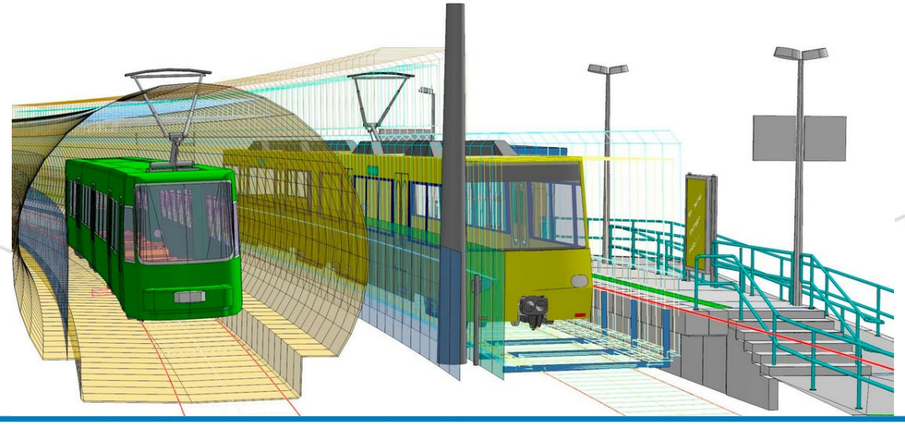




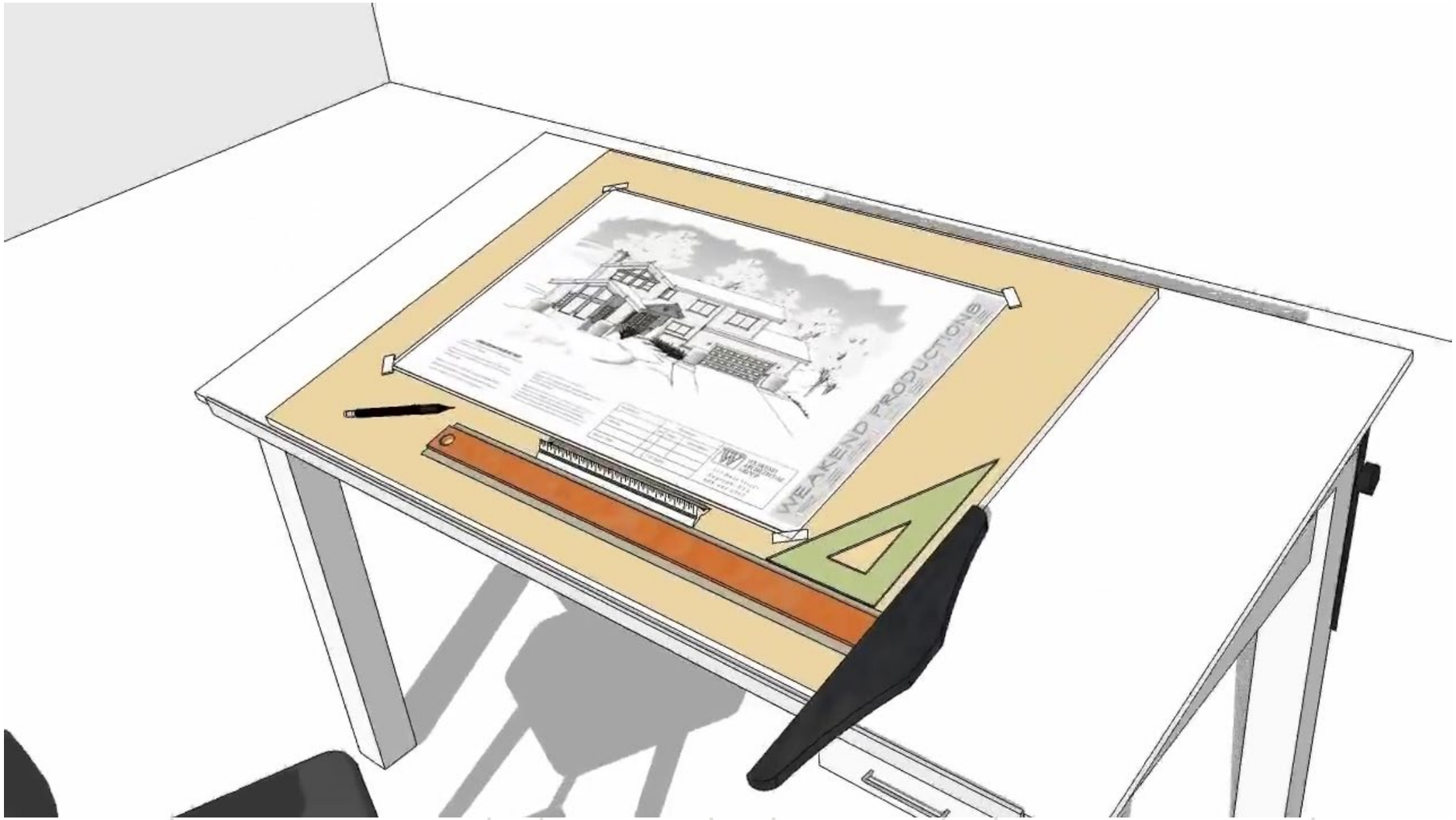
- 计算机辅助设计


bottle1.max - Autodesk 3ds Max 8 - Stand-alone License
Edit UVWs


AutoCAD 2002 - NICHT FÜR DEN WIEDERVERKAUF



- 计算机辅助设计



课程概要

- 现代计算机图形学
- 计算机图形学发展史
- 计算机图形学应用
- 如何学习计算机图形学

为什么要学习计算机图形学?

- 广泛的应用
- 庞大的市场
  - 游戏、电影、教育、科学
  - 就业机会
- 兴趣
  - 发挥创造力绘制绚丽的图像

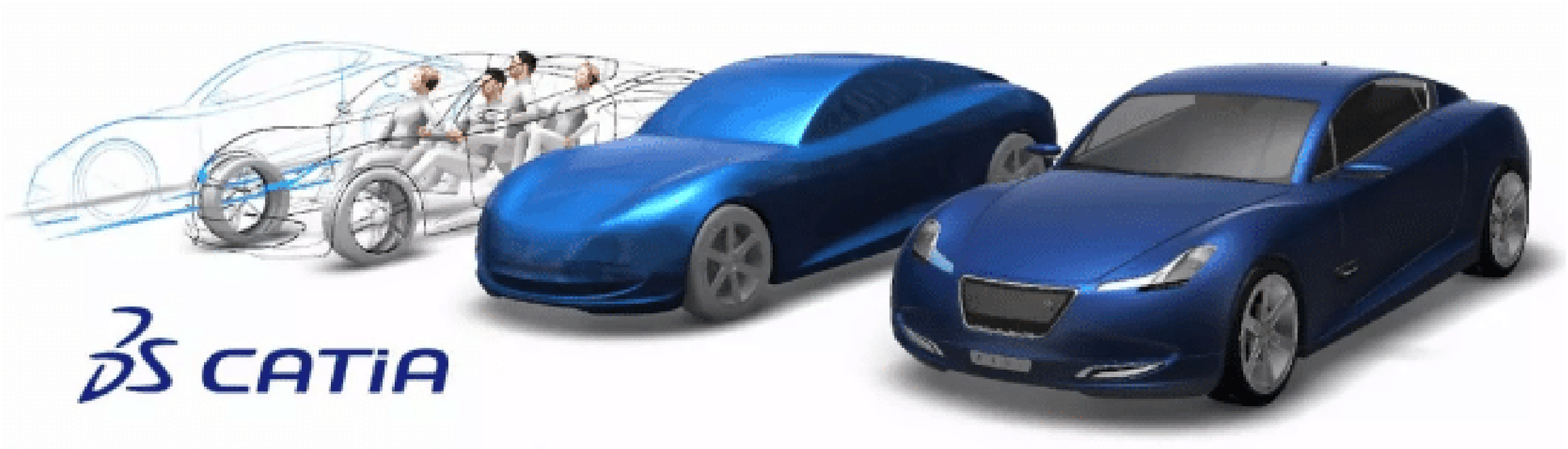
为什么要学习计算机图形学?

- 学以致用
  - Waking up every morning, thinking about the work undertaken and the technology developed which will have a tremendous impact and changes on human life, I will be extremely excited and agitated.

    – Bill Gates
  - 在美国，计算机图形学已经形成一条完整的产业链
  - 在中国，我们仍需要更多人才的参与
    - 尤其是工业设计、科学可视化等领域…

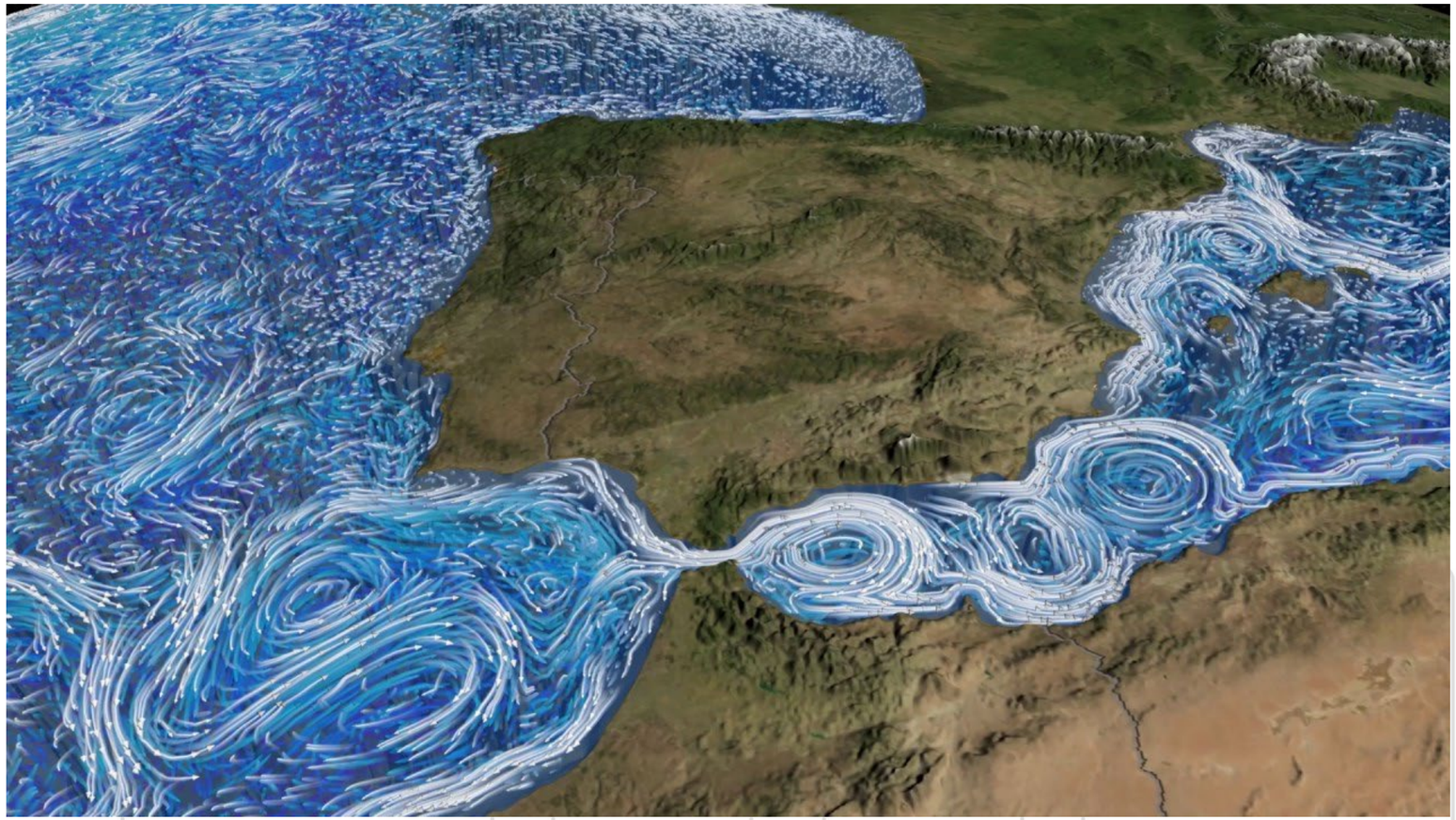
为什么要学习计算机图形学?

- 学以致用
  - 在中国，我们仍需要更多人才的参与
    - 新闻：“上海某企业用盗版CATIA软件造飞机! 被重罚了900万！”
      - 法国公司达索（飞机制造企业）起诉上海某企业
      - 达索代表作有幻影战斗机和阵风战斗
      - 我国歼10、歼20、飞豹战斗轰炸机、长征5火箭，歼11B、歼15、歼16、运20



为什么要学习计算机图形学?

- 学以致用
  - 在中国，我们仍需要更多人才的参与
    - 美国面向超级计算机的可视化系统向来不对华销售



图片来自NASA

学习计算机图形学需要什么?

- 跨学科的知识与技能
  - 数学、物理、计算机、艺术
- 好奇心
  - 对未知世界的强烈好奇心
  - 对技术的渴望与追求
- 创造力
  - 敢想敢做：不断思考、积极尝试
- 实践
  - 在实践中掌握知识：C++、线性代数、几何、优化方法…
  - 实践是掌握知识最好的方法



通过这门课我们将学习什么？

- 基础计算机图形学算法
- 三维几何处理
- 真实感渲染
  - 算法与数据结构
- OpenGL编程
  - 我们不是通过这门课学习如何使用软件设计模型渲染动画
  - 我们学习的是如何编制这些软件的基本技术
    - 需要熟练使用C++，更需要大量编程实践掌握这些技能
    - 课上内容涉及的数学知识大多较为基础
- 其他

- 国际会议
  - ACM SIGGRAPH及ACM SIGGRAPH（Asia）
    - ACM Special Interest Group on computer GRAPHics and interactive techniques
    - 1967年，由布朗大学教授Andries van Dam以及IBM纽约科学中心负责人Sam Matsa创建
    - 1974年，首届ACM SIGGRAPH在科罗拉多大学召开
    - http://www.siggraph.org



https://s2025.siggraph.org/



https://asia.siggraph.org/2025/

- ACM SIGGRAPH参会人数

2026	Los Angeles & Virtual	TBC		
2025	Vancouver & Virtual			
2024	Denver & Virtual [9]	8,800[10]	TBC	[1]
2023	Los Angeles & Virtual	14,275[11]	TBC	[2]
2022	Vancouver & Virtual[12][13]	>11,700	>90	
2021	Los Angeles & Virtual[14]	—		
2020	Virtual[15]	>10,000		
2019	Los Angeles[16]	18,700	180	
2018	Vancouver[17][18]	16,637[19]	160	
2017	Los Angeles[20]	16,500	150	
2016	Anaheim[21]	14,000	153	
2015	Los Angeles[22]	14,800	143	
2014	Vancouver[23]	14,045	175	
2013	Anaheim[24]	17,162	180	
2012	Los Angeles[25]	21,212	161	
2011	Vancouver[26]	15,872	156	
2010	Los Angeles[26]	22,549	160	
2009	New Orleans[27]	11,000	140	
2008	Los Angeles[28]	28,432	230	
2007	San Diego[29]	24,043	230	
2006	Boston[30]	19,764	230	
2005	Los Angeles[31]	29,122	250	
2004	Los Angeles[32]	27,825	229	
2003	San Diego[33]	24,332	240	
2002	San Antonio[34]	17,274	225	
2001	Los Angeles[35]	34,024	303	
2000	New Orleans[36]	25,986	316	
1999	Los Angeles[37]	42,690	337	
1998	Orlando[38]	32,210	327	
1997	Los Angeles[39]	48,700		
1996	New Orleans[40]	28,500	321	
1995	Los Angeles[40]	40,100	297	
1994	Orlando[40]	25,000	269	
1993	Anaheim[40]	27,000	285	
1992	Chicago[40]	34,148	253	
1991	Las Vegas[40]	23,100	282	
1990	Dallas[40]	24,684	248	
1989	Boston[40]	27,000	238	
1988	Atlanta[40]	19,000	249	
1987	Anaheim[40]	30,541	274	
1986	Dallas[40]	22,000	253	
1985	San Francisco[40]	27,000	254	
1984	Minneapolis[40]	20,390	218	
1983	Detroit[40]	14,000	195	
1982	Boston[40]	17,000	172	
1981	Dallas[40]	14,000	124	
1980	Seattle[40]	7,500	80	
1979	Chicago[40]	3,000	79	
1978	Atlanta[40]	1,500	44	
1977	San Jose[40]	750	38	
1976	Philadelphia[40]	300	10	
1975	Bowling Green, Ohio[40]	300		
1974	Boulder[40]	600		

截取自wikipedia

- Eurographics
  - Eurographics Symposium on Geometry Processing (SGP)
  - Eurographics Symposium on Rendering (EGSR)
  - Eurographics Symposium on Computer Animation (SCA)





计算机图形学前沿

- Pacific Graphics
  - Pacific Graphics 2025
  - Taipei
  - https://pg2025.nccu.edu.tw/



- CAD/Graphics & Chinagraph
  - CAD/Graphics 2025
  - Yantai
  - https://cadgraphics2025.sdtbu.edu.cn/



- 文章下载
  - http://kesen.realtimerendering.com/



Ke-Sen Huang's Home Page

- I got my Ph. D from the Department of Computer Science of National Tsing-Hua University, Taiwan.
- My research interests include: animation synthesis, animation summarization, and motion retrieval.
- My Web Changelog

Paper Collection / Resources

- Open Access to ACM SIGGRAPH-Sponsored Content: For both SIGGRAPH and SIGGRAPH Asia, conference content is freely accessible in the ACM Digital Library for a one-month period that begins two weeks before each conference, and ends a week after it concludes.
- Journal of Computer Graphics Techniques
- Point-based Graphics Papers
- Physics-Based Animation Resources (Maintained by Christopher Batty)
- Real-Time Rendering Resources (Maintained by Tomas Akenine-Möller, Eric Haines, and Naty Hoffman)
- Visualization Paper Collection (IEEE VisWeek papers / EuroVis papers / IEEE Pacific Vis papers) (Maintained by ZJU-VAG)

Computer Graphics Conference and Special Issue Calendar

- CFP - The Springer Encyclopedia of Computer Graphics and Games (ECGG) (PDF)
- 2013, 2012, 2011, 2010, 2009, 2008

SIGGRAPH 2025 papers on the web

Page maintained by Ke-Sen Huang. If you have additions or changes, send an e-mail.

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Note that when possible I link to the page containing the link to the actual PDF or PS of the preprint. I prefer this as it gives some context to the paper and avoids possible copyright problems with direct linking. Thus you may need to search on the page to find the actual document.

ACM Digital Library: ACM Transactions on Graphics (TOG) Volume 44, Issue 4 (July 2025) Proceedings of ACM SIGGRAPH 2025

SIG/TOG: Journal Paper for presentation at SIGGRAPH 2025

SIG: Conference Paper for presentation at SIGGRAPH 2025

TOG: Selected ACM TOG Paper for presentation at SIGGRAPH 2025

ACM Digital Library (DOI) Link for the paper **Paper Abstract** **Author Preprint** **Paper Video**

Paper Presentation **Paper Images** **Paper Data** **Demo Program or Source Code** **Related Links**

Changelog

Linear & Non-Linear AI Designs

AssetDropper: Asset Extraction via Diffusion Models with Reward-Driven Optimization
Lanjiong Li* (The Hong Kong University of Science and Technology (Guangzhou)), Guanhua Zhao* (Peking University), Lingting Zhu* (The University of Hong Kong), Zeyu Cai (The Hong Kong University of Science and Technology (Guangzhou)), Lequan Yu (The University of Hong Kong), Jian Zhang (Peking University), Zeyu Wang (The Hong Kong University of Science and Technology (Guangzhou) and The Hong Kong University of Science and Technology) * Equal Contribution

MoVer: Motion Verification for Motion Graphics Animations
Jiaju Ma, Maneesh Agrawala (Stanford University)

EditDuet: A Multi-Agent System for Video Non-Linear Editing
Marcelo Sandoval-Castaneda (Toyota Technological Institute at Chicago), Bryan Russell (Adobe), Josef Sivic (Adobe and Czech Technical University), Gregory Shakhnarovich (Toyota Technological Institute at Chicago), Fabian David Caba Heilbron (Adobe)

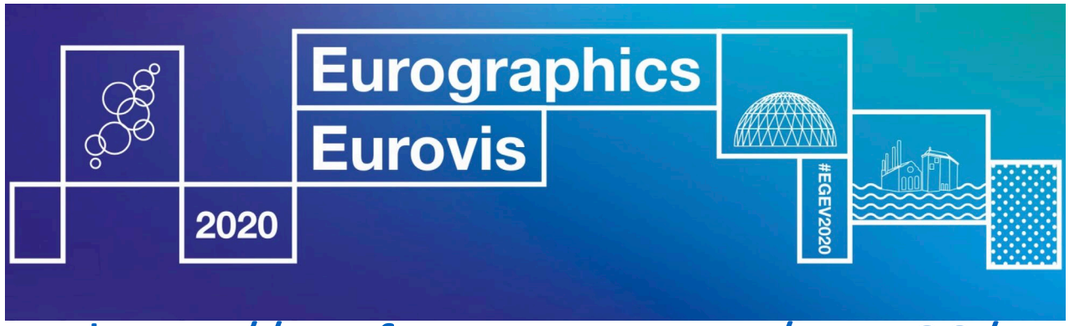
Order Matters: Learning Element Ordering for Graphic Design Generation (SIG/TOG)
Bo Yang, Ying Cao (ShanghaiTech University)

DesignManager: An Agent-Powered Copilot for Designers to Integrate AI Design Tools into Creative Workflows (SIG/TOG)

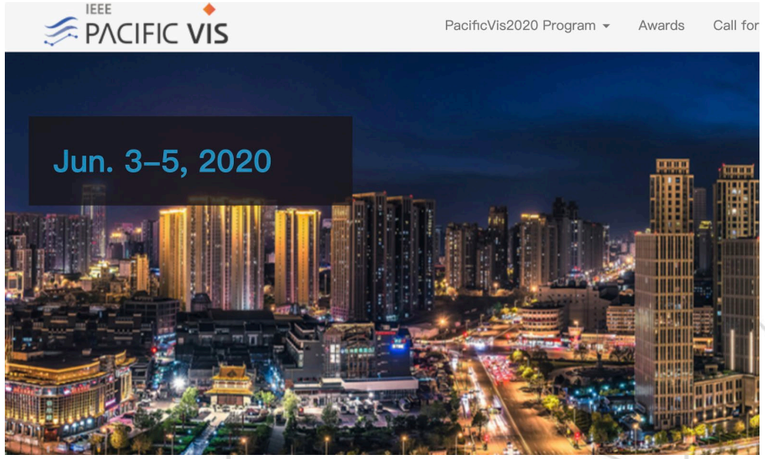
- 可视化会议
  - VIS, EuroVis, PacificVis, ChinaVis



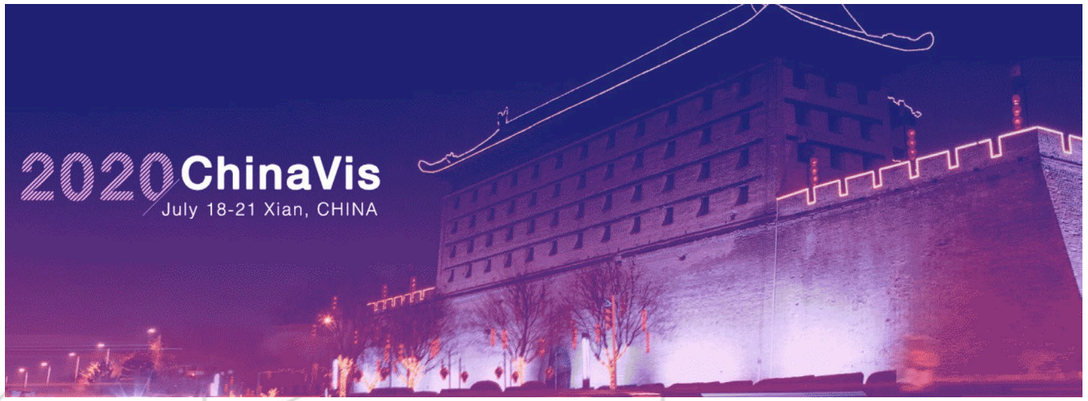
http://ieeevis.org/year/2020/welcome



https://conferences.eg.org/egev20/



http://vis.tju.edu.cn/pvis2020

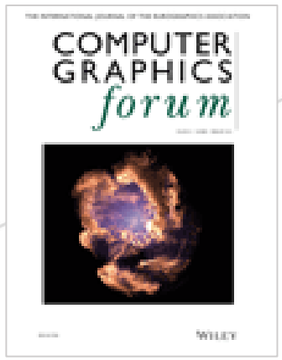


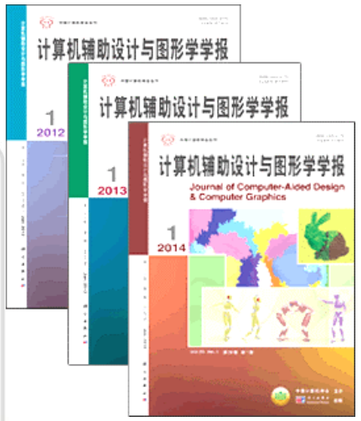
http://chinavis.org/2020/

计算机图形学前沿

- 期刊（国际）
  - ACM Transactions on Graphics
  - IEEE Transactions on Graphics and Visualization
  - Computer Graphics Forum
  - Computer Aided Geometric Design
  - Computer-aided Design
  - The Visual Computer
  - Graphical Models
  - Computer & Graphics
  - Computer Graphics & Applications
- 期刊（国内）
  - 软件学报
  - 计算机学报
  - 计算机辅助设计与图形学学报
  - 中国图象图形学报
- 会议
  - Siggraph
  - Siggraph Asia
  - Eurographics
  - Pacific Graphics
  - Symposium on Geometry Processing
  - Shape Modeling International
  - Chinagraph
  - IEEE VIS
  - EuroVis, PacificVis, ChinaVis
  - IEEE VR







其他资源

- NVIDIA GameWorks™ Samples
  - https://developer.nvidia.com/gameworks-samples-overview

NVIDIA GameWorks™ OpenGL Samples

Get the documentation or download the NVIDIA GameWorks™ OpenGL samples here:

Download ›

New Browse or Clone Source from GitHub:

GitHub ›

New samples in 2.0:

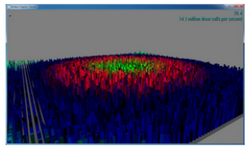
- Blended Antialiasing Sample
- Cascaded Shadow Mapping Sample
- Conservative Rasterization Sample
- Normal-Blended Decal Sample
- Weighted, Blended, Order-independent Transparency Sample

Bindless Graphics Sample

- Category: **Performance**

This sample demonstrates the large performance increase in OpenGL that is made possible by 'Bindless Graphics'. These extensions allow applications to draw large numbers of objects with only a few setup calls, rather than a few calls per object, thus reducing the driver overhead necessary to render highly populated scenery.

Docs ›

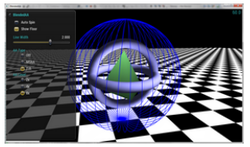


★NEW:Blended AA

- Category: **Performance, Visuals**

This sample implements a two-pass additive blending anti-aliasing technique using Target-Independent Rasterization (TIR), which should give comparable results to MSAA with a reduced memory footprint.

Docs ›

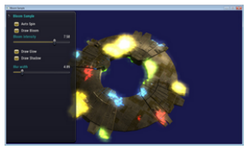


Bloom Sample

- Category: **Visuals**

This sample demonstrates creating a glow effect by post-processing the main scene. It heavily leverages FBO render targets across multiple steps/passes with custom effects processing shaders. It also integrates shadow mapping to demonstrate self-illumination cutting through the shadow effects.

Docs ›



NVIDIA GameWorks™ DirectX Samples

Get the documentation or download the NVIDIA GameWorks™ Direct3D samples here.

Download ›

New Browse or Clone Source from GitHub:

GitHub ›

New samples in 1.2:

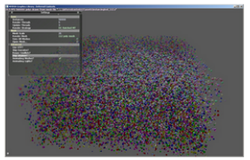
- Antialiased Deferred Rendering
- Motion Blur D3D Advanced Sample

D3D Deferred Contexts Sample

- Category: **Performance**

This sample shows how to use D3D11 Deferred Rendering contexts to lower the CPU overhead and improve performance when rendering large numbers of objects per frame, in situations where instancing is not feasible.

Docs ›

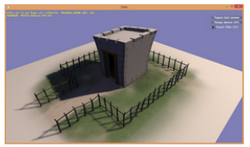


FXAA 3.11 Sample

- Category: **Performance**,**Visuals**

This sample presents a high performance and high quality screen-space software approximation to anti-aliasing called FXAA.

Docs ›

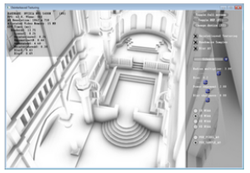


Deinterleaved Texturing Sample

- Category: **Performance**,**Visuals**

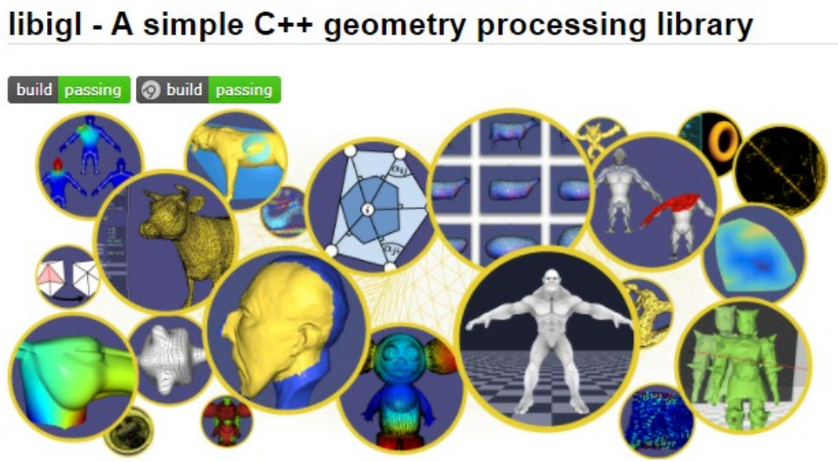
This sample demonstrates how a large, sparse and jittered post-processing filter (here a SSAO pass with a 4x4 random texture) can be made more cache-efficient by using a Deinterleaved Texturing approach.

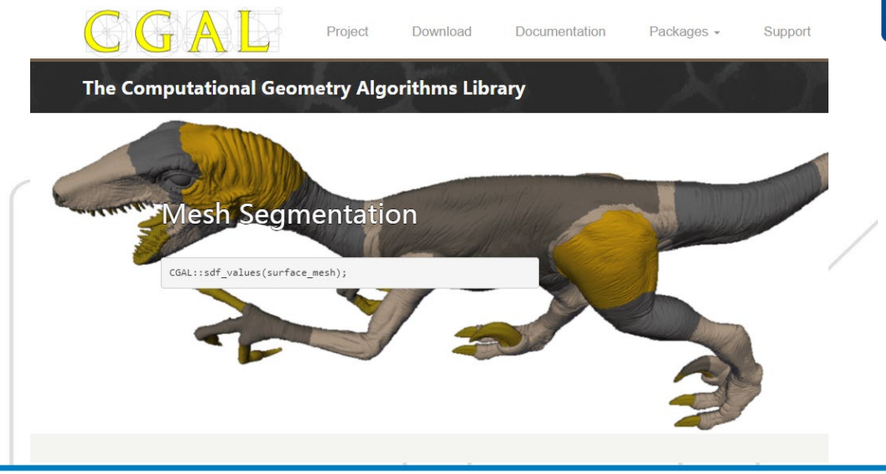
Docs ›



其他资源

- CGAL： the Computational Geometry Algorithm Library
  - https://www.cgal.org/index.html
- OpenMesh
  - https://www.openmesh.org/
- Libigl
  - https://libigl.github.io/

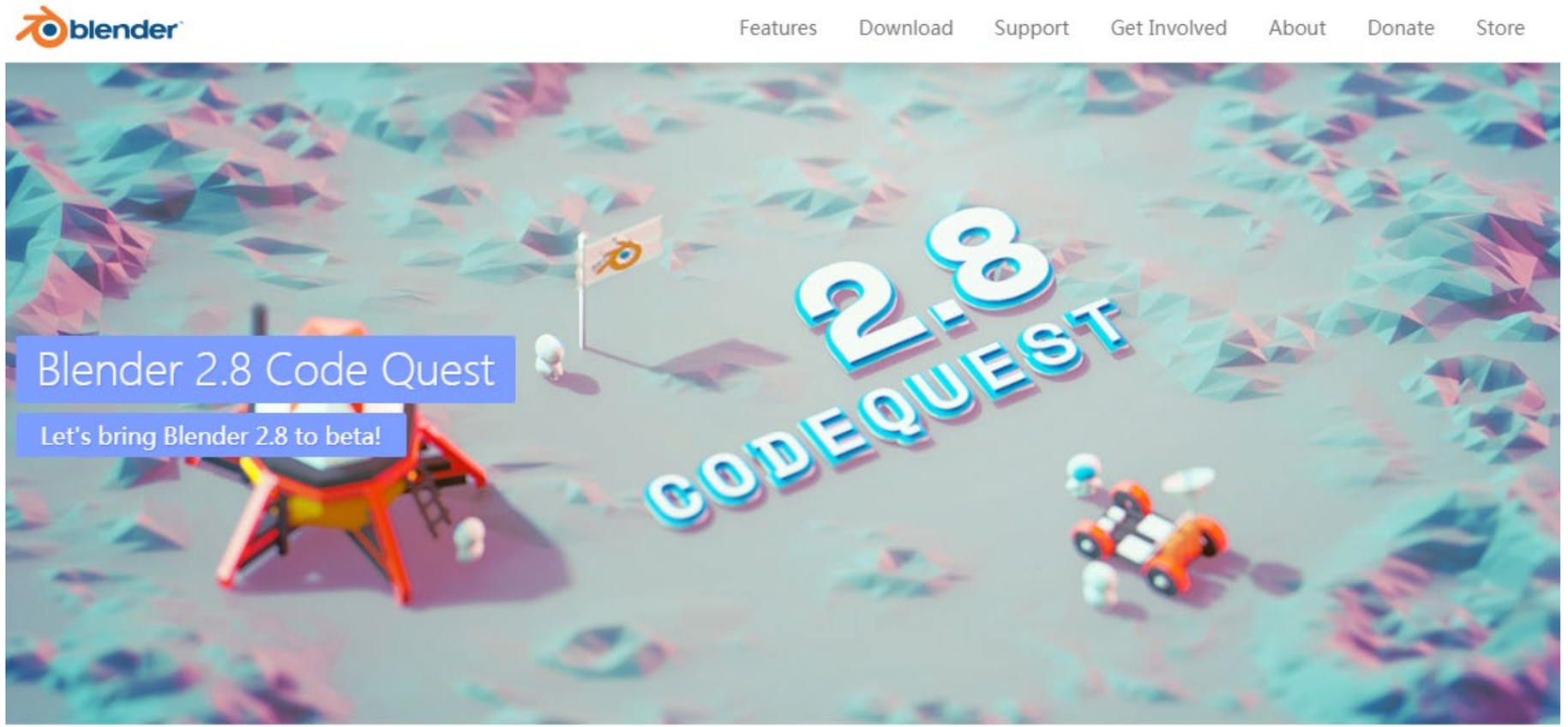






其他资源

- Blender
  - https://www.blender.org/



其他资源

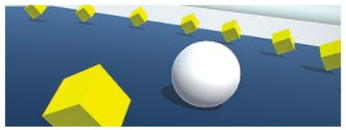
- Unity 3D
  - https://unity3d.com/cn

项目



Interactive Tutorials (4)

Get Started with Unity.



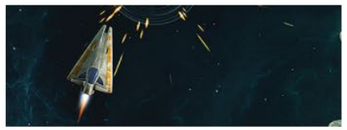
Roll-a-ball tutorial (9)

Build your first simple game and Learn to code in C#



2D Game Kit (38)

Explore the ancient and mysterious alien planet where our Principal Engineer, Ellen has crash landed.



Space Shooter tutorial (19)

Blast some Asteroids!



Survival Shooter tutorial (12)

They mostly come at night..



Tanks tutorial (8)

2-players, 1 keyboard, Tank vs Tank.



Adventure Game Tutorial (7)

Learn to create the systems used to develop an adventure game in this intermediate level project.



2D Roguelike tutorial (14)

Procedural level Survive-em-up!



Tower Defense Template (10)

Learn how to create your own Tower Defense game



2D UFO Tutorial (9)

New? Want to make 2D games? Start here.



Procedural Cave Generation tutorial (9)

Let's get spelunking.



Creating Believable Visuals (9)

In this article we look at the fundamentals of setting up good baselines for believable visuals.

Questions?

